

Synthetic Anomalies Improve Anomaly Detection: Learning Guarantees and Applications.

Tian-Yi Zhou¹, Matthew Lau², Xiangchi Yuan², Jizhou Chen², Wenke Lee² and Xiaoming Huo³

¹Data Science Institute, Columbia University in the City of New York

²School of Cybersecurity and Privacy, Georgia Institute of Technology

³H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Abstract

Anomaly detection (AD) is a critical task across domains such as cybersecurity and healthcare. In unsupervised AD, where no real anomalies are available during training, an effective principle is to train a classifier to distinguish normal data from synthetic anomalies. We extend this principle to semi-supervised AD, where a limited amount of known anomalies is available. We propose a **theoretically-grounded and empirically effective framework** for semi-supervised AD that combines known and synthetic anomalies during training. Our starting point is that unsupervised AD, semi-supervised AD, and binary classification are not separate problems but a single family, indexed by the choice of *reference density* against which normal data are compared. Placing them on this one axis makes it clear what semi-supervised AD needs: a reference density that is bounded away from zero, which is exactly what synthetic anomalies supply. We show that this choice enables (i) better anomaly modeling in low-density regions and (ii) an explicit convergence guarantee for our proposed ReLU network classifier. Our theory establishes the statistical benefits of adding synthetic anomalies and guides both how they should be generated and how many are needed. We empirically validate our framework on five diverse benchmarks, observing consistent performance gains. These improvements also extend beyond our theoretical framework to other classification-based AD methods, validating the generalizability of the synthetic anomaly principle in AD.

1 Introduction

Anomaly detection (AD) — the task of identifying data that deviate from expected behavior — is central in many domains, including detecting manufacturing defects [3], identifying harmful text [9], detecting cyber-attacks [49, 29], and diagnosing medical conditions [37, 16]. Most AD research focuses on the unsupervised setting, where only normal data are available during training, because real anomalies are often difficult to collect and label, as in the case of cyber-attacks. With recent advances in data collection, however, some real anomalies can become available for training. Many unsupervised methods do not handle this additional information and remove these real anomalies from their training data (e.g., Kim et al. [25], Qiu et al. [36], Shenkar and Wolf [46], Xiao and Fan [53]). Ideally, models should make effective use of these known anomalies to improve detection.

In this work, we build on a key principle of unsupervised AD: training a classifier to distinguish normal data from randomly generated synthetic anomalies. This approach is supported by theoretical guarantees and empirical success in unsupervised settings [48, 60, 47], but whether synthetic anomalies remain beneficial when known anomalies are also available is not yet well understood.

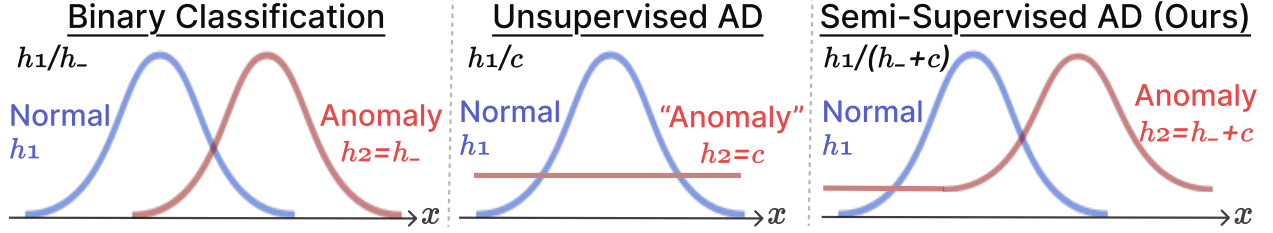


Figure 1: **Mathematical Formulations.** Different tasks based on the common test h_1/h_2 . Our semi-supervised AD formulation combines known anomaly information h_- with the density level set estimation formulation from unsupervised AD.

At first glance, adding synthetic anomalies when known anomalies are already available may seem counterproductive: mixing the two could dilute the signal provided by known anomalies, while synthetic anomalies may also fall in regions occupied by normal data. Despite these concerns, **we develop a theoretically-grounded and empirically effective framework for semi-supervised AD that combines known and synthetic anomalies during training**, demonstrating the statistical and practical benefits of this approach.

Let h_1 denote the density function of normal data. We unify unsupervised AD, semi-supervised AD, and binary classification through a common likelihood-ratio test

$$\frac{h_1}{h_2} > \rho,$$

with each problem corresponding to a different *reference density* h_2 (Figure 1). Unsupervised AD uses a constant reference density, binary classification uses the known anomaly density $h_2 = h_-$, and our semi-supervised formulation uses the mixture $h_2 = \tilde{s}h_- + (1 - \tilde{s})$. Combining known anomalies with uniformly generated synthetic anomalies during training realizes this mixture and keeps the h_2 bounded away from zero. This approach provides two benefits: it (i) enables detection in low-density regions of the normal distribution, and (ii) facilitates learning, supporting both a principled model of AD and provable convergence guarantees for neural network classifiers. Our theory also guides how many synthetic anomalies to generate, addressing concerns about dilution of the known anomaly signal and overlap with normal data.

We translate our theoretical framework into a practical implementation and evaluate it on five real-world datasets spanning cybersecurity, healthcare, manufacturing, and harmful-text detection. Our experiments show that adding synthetic anomalies improves detection of both known and unknown anomalies. These gains extend beyond our proposed model to other state-of-the-art classification-based AD methods, supporting the broader applicability of this approach.

We summarize our contributions below:

- We give a mathematical formulation for semi-supervised AD as a likelihood-ratio test against a mixture reference density, unifying it with unsupervised AD and binary classification in a single family.
- We show that adding synthetic anomalies to the anomaly class during training sidesteps two potential problems of AD practices.
- To our knowledge, we establish the first convergence rate for neural network classifiers trained with a non-representative labeled anomaly sample. Section 2 and Table 1 clarify the relationship to existing theory.

- Our experiments demonstrate that adding synthetic anomalies improves detection accuracy across different domains. This improvement extends beyond our proposed neural-network-based classifier to other classification-based AD methods, highlighting our method’s generalizability.

2 Related Works

Anomaly detection has been studied under several training settings, with terminology varying across the literature. Throughout this paper, we use “semi-supervised AD” to describe learning from normal data and a limited sample of labeled anomalies, following the bulk of the applied literature (e.g., [17, 40, 35], and more). Some studies instead use this term for learning from normal data alone [41], while Blanchard et al. [4] consider *semi-supervised novelty detection* using normal data and unlabeled, potentially contaminated observations. These settings differ in the information available during training and the assumptions needed to obtain learning guarantees.

Theory for AD: What Exists Table 1 places our work against the existing theory. Two established strands of research are particularly relevant: density level set estimation and Neyman–Pearson classification.

The first strand formulates AD as density level set estimation and solves it by classifying normal data against a reference distribution. Steinwart et al. [48] established consistency for support vector machines (SVM), Scott and Nowak [44] studied minimum volume sets, and Zhou et al. [60] obtained optimal excess risk rates for ReLU networks trained against synthetic anomalies. More recently, Hatrick and Zhou [19] showed that shallow ReLU networks suffice for consistency in the same framework. All of these works consider the unsupervised setting, where no real anomalies are available during training, and therefore do not address how to incorporate a limited sample of known anomalies.

The second strand formulates AD as a Neyman–Pearson classification problem, which minimize the probability of missing an anomaly (type II error) while keeping the probability of classifying a normal observation as anomalous (type I error) low [50]. Blanchard et al. [4] reduce semi-supervised novelty detection to Neyman–Pearson classification and obtain detectors that are consistent regardless of the novelty distribution. Kalan et al. [23] study transfer learning for outlier detection when target anomaly samples are scarce, providing learning guarantees under shifts between source and target anomaly distributions. Their analysis focuses on transferring information across anomaly distributions, while ours examines how synthetic anomalies shape the reference density and improve learning in semi-supervised AD.

Finally, Hien et al. [21] derive excess risk bounds for a semi-supervised AD method built on unbiased risk estimation, and Du et al. [12] give error bounds for out-of-distribution detection using unlabeled wild data. Both concern auxiliary data rather than the reference-density question we study.

Our Position We do not claim the first theory for anomaly detection, the first treatment of semi-supervised learning for AD, or the invention of synthetic anomalies. What Table 1 shows is empty is the combination: **a non-representative labeled anomaly sample, synthetic anomalies, ReLU network classifiers, and an explicit convergence rate**. Our contribution is to show that these settings are not separate problems but points on a single axis — the choice of reference density h_2 — and that moving along that axis is what makes the semi-supervised case learnable.

Table 1: **Where this work sits.** Training data columns: **N** normal data, **A** labeled anomalies, **U** unlabeled data, **S** synthetic anomalies drawn from a known reference density. The combination in the final row — labeled anomalies together with synthetic anomalies, for neural network classifiers, with an explicit rate — is what is unoccupied.

Work	Training data				Hypothesis class	Formulation	Guarantee
	N	A	U	S			
Steinwart et al. [48]	✓			✓	SVM	Level set / likelihood-ratio test	Consistency
Scott and Nowak [44]	✓			✓	SVM	Minimum volume set	Rate
Blanchard et al. [4]	✓		✓		Kernel	Neyman–Pearson	Consistency
Sippl [47]	✓			✓	Neural network	Level set (heuristic)	None
Hien et al. [21]	✓	✓	✓		Neural network	Risk estimation	Excess risk bound
Zhou et al. [60]	✓			✓	ReLU network	Level set / likelihood-ratio test	Rate
Du et al. [12]	✓		✓		Neural network	OOD separability	Error bound
Kalan et al. [23]	✓	✓			Neural network	Neyman–Pearson (transfer learning)	Rate
Hatrick and Zhou [19]	✓			✓	Shallow ReLU	Level set / likelihood-ratio test	Consistency
This work	✓	✓		✓	ReLU network	Level set / LRT with reference density h_2	Rate

Semi-Supervised AD in Practice Unlike unsupervised AD methods, which assume all training data are normal, a large applied literature leverages a known anomaly sample during training with considerable empirical success [17, 40, 35, 62, 27, 26, 15, 54, 11, 59]. These methods are usefully grouped by the inductive bias they impose: *geometric* (equality separation [27, 26], DevNet [35]), *statistical* (DROCC [15]), and *autoencoding* (ABC [54], DeepSAD [40]). Han et al. [17] show that even 1% labeled anomalies let such methods outperform unsupervised AD. Our work complements these heuristics by quantifying the statistical benefits of synthetic anomalies in semi-supervised AD and providing theoretical guidance on how to generate them and how many to use.

Auxiliary and Synthetic Data Previous studies have explored augmenting AD training data with synthetic anomalies generated by perturbing normal observations [13, 7, 11, 6], or with outliers drawn from external datasets [20]. A complementary line of research connects AD to classification against a reference distribution [48, 44]. Within this framework, Sippl [47] trained neural network classifiers to distinguish normal data from uniformly generated synthetic anomalies, and Zhou et al. [60] established minimax-optimal convergence rates for ReLU classifiers in the unsupervised setting. Our work extends this classification principle to semi-supervised AD, examining how synthetic anomalies can complement a limited, potentially non-representative sample of known anomalies. We develop a unified statistical formulation that explains the benefits of combining known and synthetic anomalies and yields explicit convergence guarantees.

3 Formulating Anomaly Detection

In this section, we provide a general AD formulation assuming full knowledge of anomalies. Then, we specify a formulation of semi-supervised AD that relaxes this assumption. Table 2 collects the notation used throughout; the reader who wants only the main line of argument can read h_2 as “whatever we compare normal data against” and skip the rest on a first pass.

3.1 Background: Modeling Anomaly Detection as Binary Classification

First, consider a binary classification problem between $Y = 1$ (normal class) and $Y = -1$ (anomaly class). Let μ be a known probability measure on the domain $\mathcal{X} \subseteq \mathbb{R}^d$. Without loss of generality,

Table 2: **Notation.** Symbols are grouped by role. The reference density h_2 is the object this paper is about: every formulation in Figure 1 is the same likelihood-ratio test under a different choice of it.

<i>Data and distributions</i>	
$\mathcal{X} = [0, 1]^d$	Input domain, d the ambient dimension
μ	Known reference measure on $\mathcal{X}$ (uniform, unless stated)
h_1	Density of normal data with respect to μ
h_-	Density of the known anomalies
h_2	Reference density the normal data are tested against
s	Proportion of normal data: $P(Y = 1) = s$
$\tilde{s}$	Mixture weight on known anomalies in $h_2 = \tilde{s}h_- + (1 - \tilde{s})$
ρ	Threshold of the likelihood-ratio test $h_1/h_2 > \rho$
<i>Samples</i>	
n	Number of normal training points
n^-	Number of known (labeled) anomalies
n'	Number of synthetic anomalies generated
$n_{\min}$	$\min\{n, n^-, n'\}$; the rate is governed by this
T, T^-, T'	The corresponding training sets
<i>Learning objects</i>	
$f_P(X) = E[Y X]$	Regression function
$f_c = \text{sign}(f_P)$	Bayes classifier
$R(f), R^*$	Misclassification error and its infimum; $R(f) - R^*$ is the excess risk
$S_{\mu, h_1, h_2, \rho}(f)$	AD error: mismatch between $\{f > 0\}$ and $\{h_1/h_2 > \rho\}$
f_{ERM}	Empirical risk minimizer over $\mathcal{H}_\tau$ under hinge loss
<i>Problem difficulty</i>	
α	Hölder smoothness of f_P ; larger means smoother
q	Tsybakov noise exponent; larger means better class separation
<i>Hypothesis class</i>	
σ, σ^k	ReLU and ReLU ^k activations
$\sigma_\tau, \sigma_\tau^k$	Approx-sign output activation, bandwidth $\tau > 0$
L^*, w^*	Depth and maximum width of the neural network
v^*	Number of nonzero parameters in the neural network
K^*	Bound on the magnitude of any single parameter
$\mathcal{H}_\tau$	The hypothesis space $\text{span}\{\sigma_\tau \circ f\}$

let $\mathcal{X} = [0, 1]^d$. Assume data from the normal and anomaly classes are drawn respectively from unknown distributions Q and W on $\mathcal{X}$, where Q has density h_1 with respect to μ , and W has density h_2 with respect to μ .

Let $s \in (0, 1)$ denote the proportion of normal data on $\mathcal{X}$ such that

$$P(Y = 1) = s \quad \text{and} \quad P(Y = -1) = 1 - s.$$

Let P be a probability measure on $\mathcal{X} \times \mathcal{Y}$ such that the marginal distribution on $\mathcal{X}$ is

$$P_{\mathcal{X}} = sQ + (1 - s)W.$$

For any classifier $\text{sign}(f)$ induced by a function $f : \mathcal{X} \rightarrow \mathbb{R}$, its misclassification error is given as

$$R(f) = P(\text{sign}(f(X)) \neq Y).$$

The best we can do is obtain the *Bayes classifier*, denoted by $\mathbf{f}_c$, which minimizes the misclassification error, in other words,

$$R(f_c) = R^* := \inf_{f: \mathcal{X} \rightarrow \mathbb{R} \text{ measurable}} R(f).$$

It is well-established in the literature [58] that the Bayes classifier f_c always minimizes the misclassification error and is explicitly given as

$$\mathbf{f}_c(\mathbf{X}) = \text{sign}(\mathbf{f}_P(\mathbf{X})),$$

where f_P is the regression function given by

$$f_P(X) := E[Y|X] = \frac{s \cdot h_1(X) - (1 - s) \cdot h_2(X)}{s \cdot h_1(X) + (1 - s) \cdot h_2(X)}, \quad \forall X \in \mathcal{X}. \quad (3.1)$$

Additional discussions about the Bayes classifier is given in Appendix A.

The Bayes classifier can also be defined with the likelihood ratio test as [2, 18]

$$\mathbb{1} \left(\frac{h_1(X)}{h_2(X)} \geq \rho \right) \quad (3.2)$$

for threshold $\rho = \frac{P(Y=-1)}{P(Y=1)}$. However, in AD, we set threshold ρ based on our desired type I and II errors.

We proceed to define the AD error of function $f : \mathcal{X} \rightarrow \mathbb{R}$. Define the set of data classified as normal as $\{f > 0\} := \{X : f(X) > 0\}$. Let $s = \frac{1}{1+\rho}$ and the classical Tsybakov noise condition [52, 51]

$$P_X(\{X \in \mathcal{X} : |f_P(X)| \leq t\}) \leq c_0 t^q, \quad \forall t > 0, \quad (3.3)$$

hold with some $c_0 > 0$ and noise exponent $q \in [0, \infty)$.

Then, for any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$, we extend Steinwart et al. [48] (proven in Appendix D.1) to derive a bound on the AD error

$$S_{\mu, h_1, h_2, \rho}(f) := \mu(\{f > 0\} \Delta \{h_1/h_2 > \rho\}) \leq C_q (R(f) - R^*)^{\frac{q}{q+1}}. \quad (3.4)$$

Here, Δ denotes the symmetric difference, $S_{\mu, h_1, h_2, \rho}(f)$ measures how well $\{f > 0\}$ matches the ground-truth set

$$\{h_1/h_2 > \rho\} := \{X : h_1(X)/h_2(X) > \rho\}$$

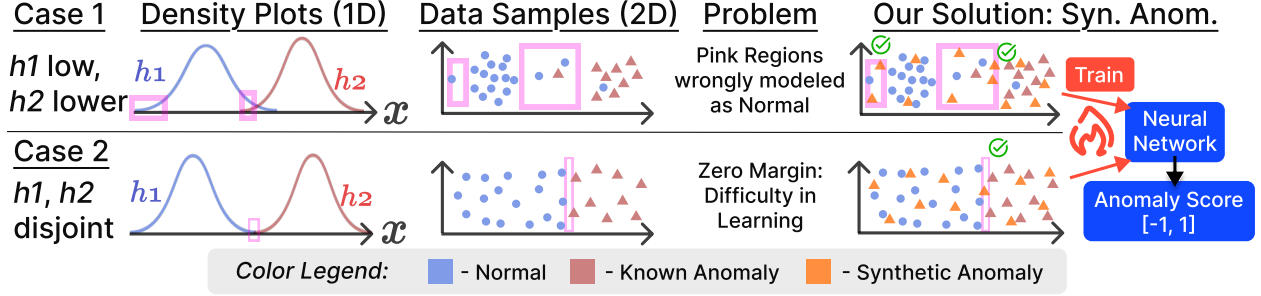


Figure 2: **Problem and Proposed Method.** We present two cases of potential issues (pink regions) from using the known anomaly density as h_2 (i.e., set $h_2 = h_-$). Sample 1D density plots and 2D data plots are provided. Circles are normal data while triangles are anomalies. The first row demonstrates the issue of wrongly modeling anomalies as normal, while the second row demonstrates the issue of insufficient regularity of learning, with the example of zero margin between normal data and known anomalies. Our solution of adding synthetic anomalies addresses these issues. In the first row, the pink region is **populated with anomalies**, so we can correctly model it as anomalous. In the second row, **synthetic anomalies smoothen the regression function**, which improve learning.

as in (3.2), whose boundary is μ -null for the continuous densities we consider. C_q is a positive constant depending on c_0 and q . From (3.4), we see $S_{\mu, h_1, h_2, \rho}(f) \rightarrow 0$ if $R(f) - R^* \rightarrow 0$. This implies that the excess risk

$$R(\cdot) - R^*,$$

a standard error metric for binary classification, serves as a surrogate for $S_{\mu, h_1, h_2, \rho}(\cdot)$ and, thus, provides a viable error metric for AD (similar to Steinwart et al. [48]). In other words, **to solve AD, we can solve a standard binary classification problem.**

However, the test-time anomaly density h_2 is not known in AD. Unsupervised AD (i.e., only normal data during training) gets around this challenge with a density level set estimation formulation [41]

$$\{h_1 \geq \rho\} := \{X : h_1(X) \geq \rho\}. \quad (3.5)$$

This formulation (3.5) can be interpreted as a likelihood ratio test between h_1 and a constant, because it is a special case of (3.2) with $h_2 \equiv 1$. In contrast, for semi-supervised AD, we would like to set h_2 to reflect our partial knowledge through our known anomaly sample.

The question we seek to answer is — **is it possible to apply this generalization to semi-supervised AD?** If so, what should h_2 be to model semi-supervised AD? Straightforwardly, we can set h_2 to be the known anomaly density. However, we proceed to show two potential issues with this approach.

3.2 Two Potential Issues without Synthetic Data

For concreteness, let our training data contain normal samples $T = \{(X_i, 1)\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} Q$ and anomalies $T^- = \{(X_i^-, -1)\}_{i=1}^{n^-} \stackrel{\text{i.i.d.}}{\sim} V$, where $V \neq W$ is an unknown distribution with density h_- . **The straightforward approach is to use T^- during training (i.e., without synthetic anomalies), implicitly setting $h_2 = h_-$.** However, we proceed to show two potential issues with this approach.

Potential Issue 1: “false negative modeling” problem The first issue is the “false negative modeling” problem, where anomalies are modeled as normal data. This may happen in regions where

normal density h_1 is low, but known anomaly density h_- is even lower, leading to $h_1(X)/h_2(X)$ exploding. In other words, **low-density regions of h_1 can still be classified as normal**. This is undesirable. Take a medical application. Refer to the density plot in the first row of Figure 2 and let x refer to blood pressure. Let h_1 refer to normal patients and h_2 (known anomalies) refer to sick patients with high blood pressure. Consider a “test-time” patient with low blood pressure X (see pink region on the left of h_1 in Figure 2). Here, $h_1(X) \gg h_2(X)$, so this patient will be modeled as normal. However, we wish to model low blood pressure as anomalous because the probability of a normal patient with low blood pressure $h_1(X)$ is low.

Potential Issue 2: “insufficient regularity of learning” problem The second potential issue is the “insufficient regularity of learning” problem, where the trained neural network classifier can produce high error. This can arise from the discontinuity of the regression function f_P , making it challenging to learn the optimal classifier. **Our observation is that, without synthetic anomalies in training data, the regression function is prone to discontinuity**, which impacts effective learning. Proposition 3.1 (proven in Appendix D.2) illustrates a general scenario (see Figure 2 and Appendix C for examples), where f_P is discontinuous despite both h_1 and h_- being continuous.

Proposition 3.1 (Separable Data with Zero Margin). *Let $r > 0$ and $\mathcal{X}$ be the union of two intersecting, closed subdomains $\mathcal{X}_1$ and $\mathcal{X}_-$ with $\text{interior}(\mathcal{X}_1 \cap \mathcal{X}_-) = \emptyset$. Suppose $h_1 \in C^r(\mathcal{X})$ has support $\mathcal{X}_1$ and $h_- \in C^r(\mathcal{X})$ has support $\mathcal{X}_-$. In particular $\mu(\mathcal{X}_1) > 0$ and $\mu(\mathcal{X}_-) > 0$, since h_1 and h_- are densities with respect to μ supported on $\mathcal{X}_1$ and $\mathcal{X}_-$ respectively. For $h_2 = h_-$, the regression function reduces to*

$$f_P(X) = \begin{cases} \frac{h_1(X)}{h_1(X)} = 1, & \text{if } X \in \text{interior}(\mathcal{X}_1), \\ -\frac{h_-(X)}{h_-(X)} = -1, & \text{if } X \in \text{interior}(\mathcal{X}_-), \end{cases}$$

which is discontinuous on $\mathcal{X}$. Moreover, for any continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$, the approximation error is at least $\|f - f_P\|_{L^\infty[0,1]^d} \geq 1$.

Next, we show that the discontinuity of f_P poses a difficulty for classification by neural networks. We consider the classical ReLU neural networks. We outline notation below.

Definition 3.2. Let $\sigma(x) = \max\{0, x\}$ be the ReLU activation function. A ReLU network $f : \mathcal{X} \rightarrow \mathbb{R}$ with $L \in \mathbb{N}$ hidden layers and width vector $\mathbf{p} = (p_1, \dots, p_L) \in \mathbb{N}^L$, which indicates the width in each hidden layer, is defined in the following compositional form:

$$f(X) = a \cdot \sigma(W^{(L)} \sigma(\dots \sigma(W^{(1)} X + b^{(1)}) \dots) + b^{(L)}) + b, \quad (3.6)$$

where $X \in \mathcal{X} = [0, 1]^d$ is the input, $W^{(i)}$ is a $p_i \times p_{i-1}$ weight matrix with $p_0 = d$ and $b^{(i)} \in \mathbb{R}^{p_i}$ is a bias vector, for $i = 1, \dots, L$, and the outer layer consists of the weight $a \in \mathbb{R}^{p_L}$ and the scalar bias $b \in \mathbb{R}$. The activation σ acts coordinatewise, so each of the L hidden layers carries its own bias $b^{(i)}$; only the output bias b is a scalar, f being real-valued. Let σ^k be the ReLU^k function, a generalization of ReLU for $k \in \mathbb{N}$, defined by $\sigma^k(x) = (\max\{0, x\})^k$. Define the “approx-sign function” $\sigma_\tau : \mathbb{R} \rightarrow [-1, 1]$, with a bandwidth parameter $\tau > 0$, as

$$\sigma_\tau(x) = \frac{1}{\tau} \sigma(x) - \frac{1}{\tau} \sigma(x - \tau) - \frac{1}{\tau} \sigma(-x) + \frac{1}{\tau} \sigma(-x - \tau). \quad (3.7)$$

We also define the generalized approx-sign^k function as $\sigma_\tau^k(x) := \frac{1}{k! \tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(x - j\tau) - \frac{1}{k! \tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(-x - j\tau)$ for $k \in \mathbb{N}$. Here, the approx-sign function is designed to approximate

the sign function (as $\tau \rightarrow 0$). Concretely, σ_τ is the piecewise-linear function built from four ReLU units that equals -1 below $-\tau$, equals $+1$ above τ , and interpolates linearly in between; equivalently, $\sigma_\tau(x) = \text{clip}(x/\tau, -1, 1)$. The bandwidth τ therefore controls the width of the transition region: as $\tau \rightarrow 0$ the function converges pointwise to $\text{sign}(x)$ away from the origin, while any $\tau > 0$ keeps it inside the ReLU network class, which is what makes the analysis tractable. Meanwhile, $k \in \mathbb{N}$ is a parameter controlling the smoothness of ReLU (and the approx-sign activation function), generalizing our analysis beyond the original non-smooth ReLU function. We defer discussions and visualizations to Appendix B.2.

The following theorem presents an upper bound for the excess risk of a function $\sigma_\tau^k(f)$ induced by the output activation σ_τ^k in terms of the bandwidth τ and the approximation error.

Theorem 3.3. *Assume the Tsybakov noise condition (3.3) holds for some exponent $q \in [0, \infty)$ and constant $c_0 > 0$. For any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$, there holds*

$$\underbrace{R(\sigma_\tau^k(f)) - R(f_c)}_{\text{excess risk}} \leq 4c_0 \left(k\tau + \underbrace{\|f - f_P\|_{L^\infty[0,1]^d}}_{\text{approximation error}} \right)^{q+1}.$$

Theorem 3.3 shows that the smaller the approximation error¹, the smaller the excess risk in classification. We discuss the significance of this theorem in Appendix B.3 and prove it in Appendix E. The proof is built on the error decomposition framework, followed by bounding and balancing the approximation and estimation errors.

From Proposition 3.1 and Theorem 3.3, we can see that **if the regression function is discontinuous, the approximation error is high (at least 1), which may lead to vacuous excess risk bounds**² (i.e., excess risk can be high and is not guaranteed to converge). Lacking theoretical guarantees, **the Bayes classifier cannot be effectively learned**. Due to (i) an undesirable formulation and (ii) lack of theoretical guarantees, we see that $h_2 = h_-$ is not ideal.

In the next section, we propose a semi-supervised AD method to mitigate these two issues.

4 Our Proposed Method: Semi-Supervised AD with Synthetic Anomalies

4.1 Overview of Our Method

Building on the previous classification framework and inspired by the connection between density level set estimation and synthetic anomalies, we propose to add synthetic anomalies to mitigate the two aforementioned issues (Figure 2). In addition to samples T and T^- , we generate a set of synthetic anomalies $T' = \{(X'_i, -1)\}_{i=1}^{n'}$, where each X'_i is sampled i.i.d. from $\mu = \text{Uniform}(\mathcal{X})$. Our full training dataset becomes $T \cup T^- \cup T'$, which we use to train a ReLU network classifier.

4.2 Mitigating Issue 1: False Negative Modeling Problem

Let $\tilde{s} \in (0, 1)$ denote a mixture parameter. By introducing synthetic anomalies, we are implicitly changing the density function representing the anomaly class to

$$h_2 = \tilde{s}h_- + (1 - \tilde{s}), \quad (4.1)$$

¹The approximation error here describes the error of function f in approximating the regression function.

²The bound could be vacuous because the excess risk is always at most 1, so an upper bound of 1 or more constrains nothing.

which corresponds to a mixture. Here, a proportion $\tilde{s}$ of anomalies are drawn from known anomaly density h_- , and the remaining proportion $(1 - \tilde{s})$ of (synthetic) anomalies are drawn from the distribution μ . We see that (in (4.1)) **h_2 is bounded away from 0** due to the constant term $1 - \tilde{s} > 0$, **preventing h_1/h_2 from exploding even when h_1 is small**. Hence, **low probability density normal data will not be modeled as anomalous** even in regions where h_2 is small.

4.3 Mitigating Issue 2: Insufficient Regularity of Learning Problem

Adding synthetic anomalies from the uniform distribution can also improve the smoothness of the regression function. Later in Section 4.5, we use this fact to show how we can effectively learn the Bayes classifier. While Proposition 3.1 illustrated that the regression function f_P can be discontinuous despite h_1 and h_- being continuous, **our next result shows that adding synthetic anomalies ensures continuity of regression function f_P** under the same conditions.

Proposition 4.4 (Synthetic Anomalies Restore Continuity). *Suppose the condition stated in Proposition 3.1 holds. If we add synthetic anomalies from $\mu = \text{Uniform}(\mathcal{X})$ (i.e., $h_2 = \tilde{s}h_- + (1 - \tilde{s})$ with $\tilde{s} \in (0, 1)$), the regression function is*

$$f_P(X) = \frac{s \cdot h_1(X) - (1 - s)\tilde{s} \cdot h_-(X) - (1 - s)(1 - \tilde{s})}{s \cdot h_1(X) + (1 - s)\tilde{s} \cdot h_-(X) + (1 - s)(1 - \tilde{s})},$$

which is C^r continuous.

For concreteness, we present two examples in Appendix C to illustrate how synthetic anomalies enhance the smoothness of regression function f_P .

Proposition 4.4 is qualitative: it says that continuity is restored, but not by how much. The following sharpening makes the gain explicit, and it is what lets the mixture weight $\tilde{s}$ be chosen rather than assumed.

Proposition 4.5 (Explicit regularity gain). *Let h_1 and h_- be α -Hölder on $\mathcal{X}$ with seminorms $[h_1]_\alpha$ and $[h_-]_\alpha$, and let $h_2 = \tilde{s}h_- + (1 - \tilde{s})$ with $\tilde{s} \in [0, 1]$. Then f_P is α -Hölder with*

$$[f_P]_\alpha \leq \frac{[A]_\alpha + [B]_\alpha}{\inf_{\mathcal{X}} B} \leq \frac{2(s[h_1]_\alpha + (1 - s)\tilde{s}[h_-]_\alpha)}{(1 - s)(1 - \tilde{s})}, \quad (4.2)$$

where $A = sh_1 - (1 - s)h_2$ and $B = sh_1 + (1 - s)h_2$, so that $f_P = A/B$.

Three things follow, and the third is the reason for stating it. First, the bound is **sharp in the regime that matters**: in the disjoint-support setting of Proposition 3.1 it is attained to within 0.4% at $\tilde{s} = 0$. Second, it **recovers Proposition 3.1** as $\tilde{s} \rightarrow 1$ — the constant diverges exactly as the synthetic anomalies are removed, which is the quantitative form of the discontinuity that Proposition 3.1 exhibits. Third, because the Hölder radius enters the approximation error linearly and Theorem 3.3 raises it to the power $q + 1$, **the excess risk of Theorem 4.11 carries an explicit factor $(1 - \tilde{s})^{-(q+1)}$** . The choice of how many synthetic anomalies to add is therefore not a free parameter: it trades this factor against the information carried by the known anomalies, a trade-off we analyze in follow-up work [61].

Remark 4.6 (When the gain is real). *The divergence in (4.2) as $\tilde{s} \rightarrow 1$ comes from $\inf B$, which vanishes only if h_- itself vanishes somewhere on $\mathcal{X}$. If instead h_- is bounded below, f_P is already α -Hölder without any synthetic anomalies and the gain is a constant factor rather than a qualitative change. This is the same boundary as Remark 4.7: synthetic anomalies buy regularity precisely when*

the known anomaly density has gaps — which is the case Proposition 3.1 describes and, we would argue, the case of interest, since a known anomaly sample that covers the whole domain is not a semi-supervised AD problem.

Remark 4.7 (Does populating the margin cost accuracy?). *Synthetic anomalies land everywhere in $\mathcal{X}$, including between the normal and known-anomaly regions, which invites the objection that we contaminate a margin we could otherwise exploit. The answer depends on what is being estimated. For binary classification between normal data and known anomalies, a wide margin is an asset and filling it would indeed hurt. But our target is the level set $\{h_1/h_2 > \rho\}$, and changing h_2 changes the target rather than corrupting the estimate of a fixed one: the boundary is pulled in toward the normal data, as in one-class methods, instead of resting midway between the two observed classes. That is the behavior we want, because test-time anomalies are not drawn from h_- . Two regimes are worth separating. When the margin has zero probability mass (Proposition 3.1), there is no margin to lose and the gain in regularity is free. When the classes are separated by a genuine margin of width $\epsilon > 0$, f_P is already continuous there, so synthetic anomalies buy no smoothness and the trade is real: known-anomaly accuracy can fall while unknown-anomaly accuracy rises. The mixture weight $\tilde{s}$ is the dial governing this trade, and our ablations over n' measure it directly (Table 5); consistent with the account above, we observe known-anomaly performance degrading only once n' grows large.*

Previously, from Proposition 3.1, we know that if f_P is discontinuous, no ReLU neural network can approximate it well. Conversely, if f_P is **continuous**, a well-established body of research has proved that ReLU neural networks can approximate it to any desired accuracy (e.g., Theorems 1 and 2 in Yarotsky [56], Theorem 5 in Schmidt-Hieber [42], Theorem 1.1 in Shen et al. [45]). However, we **cannot directly use existing results because the i.i.d. assumption is violated** — anomalies are not drawn i.i.d. from h_2 , but they are drawn from h_- (known anomalies) and μ (synthetic anomalies) separately. We proceed to derive a theoretical result that accommodates this non-i.i.d. setting.

4.4 Which Generator? Uniform Minimizes the Regularity Constant

Nothing so far requires the synthetic anomalies to be uniform. Replacing μ by an arbitrary generator density g gives $h_2 = \tilde{s}h_- + (1 - \tilde{s})g$, and Proposition 4.5 goes through verbatim with $(1 - \tilde{s})$ replaced by $(1 - \tilde{s}) \text{ess inf } g$ in the denominator and $(1 - s)(1 - \tilde{s})[g]_\alpha$ added inside the bracket in the numerator. So the theory accommodates **any** generator whose density is bounded away from zero and α -Hölder — a strictly larger class than uniform, and one worth stating explicitly, since it is the condition a practitioner must check.

Among that class, however, uniform is not an arbitrary choice.

Proposition 4.8 (Uniform minimizes the regularity constant). *Let g be any probability density on $\mathcal{X}$ with respect to μ . Then*

$$\text{ess inf}_{\mathcal{X}} g \leq 1 \quad \text{with equality iff } g \equiv 1 \text{ } \mu\text{-a.e.}, \quad [g]_\alpha \geq 0 \quad \text{with equality iff } g \text{ is constant.}$$

Consequently the uniform density simultaneously maximizes the denominator and minimizes the numerator of the bound (4.2), and is the unique minimizer of that bound over all admissible generators.

Proof. Since μ is a probability measure and $\int_{\mathcal{X}} g d\mu = 1$, we have $\text{ess inf } g \leq \int g d\mu = 1$, with equality only if g is μ -a.e. equal to its essential infimum, i.e. constant, hence $\equiv 1$. The seminorm statement is immediate. $\square$

This is a statement about the constant, not about accuracy. Proposition 4.8 identifies the unique minimizer of the *upper bound* (4.2). A generator with a larger regularity constant may still learn as well or better. What the proposition does settle is the natural objection — why not a more informative or adaptive generator? — at the level of the analysis: any generator that concentrates mass somewhere, Gaussian or learned, necessarily has $\text{ess inf } g < 1$ and pays for it in the regularity constant, and the sharper it concentrates the more it pays. The intuition is that the synthetic anomalies are not there to resemble anomalies; they are there to keep the reference density away from zero, and spreading mass as evenly as possible is exactly the way to do that. Empirically the choice matters little either way: substituting Gaussian, truncated-Gaussian and low-discrepancy generators moves pooled AUROC by at most 0.009 (Table ??). Uniform is therefore the choice the analysis prefers and one the experiments do not argue against, which is the honest form of the claim.

4.5 Proposed Neural Network with Synthetic Anomalies and Theoretical Guarantees

We proceed to establish an explicit convergence rate for the excess risk, and consequently for the AD error metric. To our knowledge this is **the first convergence rate for neural network classifiers trained with a non-representative labeled anomaly sample**.

We adopt ReLU neural networks. We construct a specific class of ReLU neural networks (i.e., our hypothesis space) to learn the Bayes classifier f_c well. We introduce some notation to formally define this hypothesis space.

Definition 4.9. Let $\|W^{(i)}\|_0$ and $|b^{(i)}|_0$ denote the number of nonzero entries of $W^{(i)}$ and $b^{(i)}$ in the i -th hidden layer, $\|\mathbf{p}\|_\infty$ denote the maximum number of nodes among all hidden layers, and $\|\boldsymbol{\theta}\|_\infty$ denote the largest absolute value of entries of the full parameter vector $\boldsymbol{\theta} = (\{W^{(i)}, b^{(i)}\}_{i=1}^L, a, b)$ of (3.6), outer layer included. For $L, w, v, K > 0$, we denote the form of neural network we consider in this work by

$$\mathcal{F}(L, w, v, K) := \{f \text{ of the form of (3.6)} : \|\mathbf{p}\|_\infty \leq w, \|a\|_0 + |b|_0 + \sum_{i=1}^L (\|W^{(i)}\|_0 + |b^{(i)}|_0) \leq v, \|\boldsymbol{\theta}\|_\infty \leq K\}.$$

With σ_τ given in (3.7), we **define our hypothesis space $\mathcal{H}_\tau$ with $\tau \in (0, 1]$ to be functions generated by $\mathcal{H}_\tau := \text{span}\{\sigma_\tau \circ f : f \in \mathcal{F}(L^*, w^*, v^*, K^*)\}$ for specific $L^*, w^*, v^*, K^* > 0$.**

Definition 4.10. To make computation feasible, it is common to adopt some convex, continuous loss to replace the 0-1 classification loss function. Among all functions in $\mathcal{H}_\tau$, we specifically consider the empirical risk minimizer (ERM) w.r.t. Hinge loss $\phi(x) := \max\{0, 1 - x\}$ defined as

$$f_{\text{ERM}} := \arg \min_{f \in \mathcal{H}_\tau} \varepsilon_{T, T^-, T'}(f), \quad (4.3)$$

where the empirical risk w.r.t. ϕ is

$$\varepsilon_{T, T^-, T'}(f) := \frac{s}{n} \sum_{i=1}^n \phi(f(X_i)) + \frac{(1-s)\tilde{s}}{n^-} \sum_{i=1}^{n^-} \phi(-f(X_i^-)) + \frac{(1-s)(1-\tilde{s})}{n'} \sum_{i=1}^{n'} \phi(-f(X'_i)), \quad (4.4)$$

which uses normal data, known anomalies and synthetic anomalies from a uniform distribution. Note that n and n^- denote the number of normal and (real) anomalous training samples respectively, and n' denotes the number of synthetic anomalies we generate.

Before the formal statement, here is what it says with the bookkeeping removed.

Theorem 4.11 (informal). Train a ReLU network to separate normal data from the union of known and synthetic anomalies. If the regression function is α -smooth and the classes separate at Tsybakov exponent q , then in ambient dimension d the excess risk decays as

$$R(\text{sign}(f_{\text{ERM}})) - R(f_c) = \tilde{\mathcal{O}}\left(n_{\min}^{-\frac{\alpha(q+1)}{d+\alpha(q+2)}}\right), \quad n_{\min} = \min\{n, n^-, n'\}.$$

Three readings matter. (i) Only the *smallest* of the three sample sizes appears, so whichever of normal data, known anomalies, or synthetic anomalies is scarcest is what limits accuracy. This holds provided the three are polynomially comparable, $\max\{n, n^-, n'\} \leq (n_{\min})^c$ for a fixed c : the largest sample size does enter the explicit bound, inside a logarithm, and it is that hypothesis which makes the logarithm harmless. (ii) Smoother f_P (larger α) and better separation (larger q) both speed convergence, and dimension slows it — the usual nonparametric picture. (iii) The rate exists at all only because synthetic anomalies make f_P continuous (Proposition 4.4); without them the approximation error need not vanish.

Formally, the excess risk of the ERM, f_{ERM} (4.3), trained on normal data, known anomalies and synthetic anomalies, converges to 0 at an explicit polynomial rate (up to a logarithmic factor) as the number of training data increases.

Theorem 4.11. Let $n, n^-, n' \geq 3, n_{\min} = \min\{n, n^-, n'\}, d \in \mathbb{N}, \alpha > 0$. Assume the Tsybakov noise condition (3.3) holds for noise exponent $q \in [0, \infty)$ and constant $c_0 > 0$, and the regression function f_P is α -Hölder continuous. Assume further that the three sample sizes are polynomially comparable: $\max\{n, n^-, n'\} \leq (n_{\min})^c$ for some fixed $c \geq 1$. Consider the hypothesis space $\mathcal{H}_\tau$ with $N = \left\lceil \left(\frac{n_{\min}}{(\log(n_{\min}))^4} \right)^{\frac{d}{d+\alpha(q+2)}} \right\rceil$, $\tau = N^{-\frac{\alpha}{d}}, K^* = 1$, and L^*, w^*, v^* depending on N, α, d given explicitly in Appendix F. For any $0 < \delta < 1$, with probability $1 - \delta$, there holds,

$$R(\text{sign}(f_{\text{ERM}})) - R(f_c) = \mathcal{O} \left(\log \left(\frac{6}{\delta} \right) \log(n_{\min}) \left(\frac{(\log n_{\min})^4}{n_{\min}} \right)^{\frac{\alpha(q+1)}{d+\alpha(q+2)}} \right) = \tilde{\mathcal{O}} \left(n_{\min}^{-\frac{\alpha(q+1)}{d+\alpha(q+2)}} \right),$$

where the implied constant depends on c but not on n, n^-, n' or δ .

The comparability hypothesis is what makes the rate a statement about $n_{\min}$ alone. The explicit bound proved in Appendix F carries a factor $\log(\frac{6}{\delta} \max\{n, n^-, n'\})$, which is controlled by the smallest sample size only if the largest is polynomially bounded by it; under $\max\{n, n^-, n'\} \leq (n_{\min})^c$ that factor is at most $(1+c) \log(\frac{6}{\delta}) \log(n_{\min})$, and the displayed form follows. The hypothesis is mild — it permits any polynomial imbalance between the three samples, and our own prescription $n' = n + n^-$ satisfies it as soon as n and n^- are themselves polynomially comparable, since then $n_{\min} = \min\{n, n^-\}$ and $n' \leq 2 \max\{n, n^-\}$ — and it is the same hypothesis under which the refined rate of Zhou et al. [61] is stated.

Theorem 4.11 tells us that when $n_{\min} = \min\{n, n^-, n'\}$ increases, the excess risk converges to 0 at a rate $\mathcal{O} \left((n_{\min})^{-\frac{\alpha(q+1)}{d+\alpha(q+2)}} \right)$ (dropping the logarithmic factor). Here $n_{\min}$ captures the minimum sample size across normal, anomalous and synthetic training data.

Remark 4.12 (On the exponent being smaller than $1/2$). The exponent $\frac{\alpha(q+1)}{d+\alpha(q+2)}$ falls below $1/2$ whenever d is large relative to α , which is slower than the $\mathcal{O}(n^{-1/2})$ familiar from parametric problems. Slow exponents are in part the price of nonparametric estimation, and a substantial literature is

devoted to attaining the best possible one [24]; but part of the slowness here is ours rather than the problem's, as Remark ?? makes precise. We report the rate in terms of d , α and q precisely so that this dependence is visible rather than buried in a constant. Nor is $c = \alpha q$ necessarily small: α grows with the smoothness of f_P — which, by Proposition 4.4, is exactly what synthetic anomalies improve — and q grows with class separation, diverging in the separable case.

In fact the crossing point is exact. Since

$$\frac{\alpha(q+1)}{d+\alpha(q+2)} > \frac{1}{2} \iff 2\alpha(q+1) > d + \alpha(q+2) \iff \alpha q > d,$$

the rate is faster than $n^{-1/2}$ precisely when $\alpha q > d$: the product of smoothness and noise exponent must exceed the ambient dimension. This identifies exactly which problems our bound treats as “easy” in the sense of admitting parametric-like rates. Whether the $n_{\min}$ dependence in particular is unimprovable is a separate question, which we take up in follow-up work [61].

Corollary 4.13 (Separable case). *As $q \rightarrow \infty$ — the separable regime, in which the normal and anomalous classes carry no probability mass near the decision boundary — the exponent satisfies*

$$\frac{\alpha(q+1)}{d+\alpha(q+2)} \longrightarrow 1,$$

so the excess risk approaches the parametric rate $\mathcal{O}(n_{\min}^{-1})$, free of the curse of dimensionality.

Concretely, at $\alpha = 1$ and $d = 10$ the exponent is 0.500 at $q = 10$, 0.902 at $q = 100$ and 0.989 at $q = 1000$. This is the behavior one should expect: when the two classes are cleanly separated, locating the boundary is an essentially parametric problem, and the nonparametric penalty in d is paid only to resolve a genuinely noisy boundary. Applying Theorem 4.11 to (3.4), we obtain with probability $1 - \delta$, the AD error $S_{\mu, h_1, h_2, \rho}(\text{sign}(f_{\text{ERM}})) = \mathcal{O}\left((n_{\min})^{-\frac{\alpha q}{d+\alpha(q+2)}}\right)$. As the number of training data grows, the AD error converges to 0, **suggesting that ReLU network can solve semi-supervised AD effectively.**

What the theory prescribes. Theorem 4.11 is not decorative: each of its parameters fixes a concrete implementation choice, and every one is checked empirically in Section 5.

- **How many synthetic anomalies.** The rate depends only on $n_{\min} = \min\{n, n^-, n'\}$, so raising n' past $\min\{n, n^-\}$ cannot improve it. We set $n' = n + n^-$, which is the natural choice that keeps n' off the binding constraint without requiring us to know which of n and n^- is the smaller; the least sufficient choice is $\min\{n, n^-\}$ itself. Larger n' is not wrong, merely wasted — and empirically it begins to dilute the known-anomaly signal (Table 5).
- **How to generate them.** The proof uses only that h_2 is bounded away from zero on $\mathcal{X}$ and α -Hölder, so the rate is the same for every generator meeting those two conditions. Among them the bound’s constant is smallest for uniform, uniquely so (Proposition 4.8); empirically the four generators we tried are within 0.009 AUROC (Table ??).
- **Network size.** N , and hence depth L^* and width w^* , are set by $n_{\min}, \alpha, d$ as given in Appendix F: capacity should grow with the scarcest sample, not with the largest.
- **Output bandwidth.** $\tau = N^{-\alpha/d}$ ties the sharpness of the approx-sign layer to the same budget, so the decision boundary is resolved no more finely than the data support.
- **What is *not* prescribed.** The theorem says nothing about optimiser, learning rate, or batch size; those we tune by structural risk minimization (Appendix G.5) and do not claim theoretical backing for.

Remark 4.14 (Which constraints bind at realistic sample sizes). *The prescriptions above are asymptotic, and it is worth being explicit about what that costs, because two of the four constraints turn out to be vacuous in the regimes we actually study.*

Instantiating the explicit constants of Appendix F at our five datasets, with $\alpha = q = 1$, gives the following. The sparsity budget $v^ = 141(d + \alpha + 1)^{3+d}N(\ell + 6)$ carries a $(d + \alpha + 1)^{3+d}$ factor: at $d = 119$ it is of order 10^{259} , against a trained network with roughly $10^{5.7}$ parameters. Here $\ell \in \mathbb{N}$ is the approximation parameter of Lemma F.26, which sets how finely the construction resolves the target: the approximation error carries a term $N2^{-\ell}$, and the proof takes $\ell = \lceil (1 + \alpha/d) \log_2 N \rceil$, the smallest choice making that term no larger than $N^{-\alpha/d}$. It enters L^* and v^* and nothing else. (It is written m in Schmidt-Hieber [42]; we reserve m for the smoothness index of C^m .) As a constraint it is therefore never active, and an experiment reporting that it has “enforced the theory’s sparsity budget” would not in fact be constraining anything. Similarly, Lemma F.26 requires $N \geq \max\{(\alpha + 1)^d, (r + 1)e^d\}$, which at $d = 119$ means $N \gtrsim 10^{52}$; the theorem’s own choice of N at $n_{\min} = 56$ is $N = 1$. Entering the regime where the construction applies would require $n_{\min}$ of order 10^{53} for NSL-KDD and 10^{446} for MVTec.*

This is not a defect peculiar to our result — constants of this shape are standard in the non-parametric neural network literature, and Theorem 4.11 is a statement about rates, which is what it claims to be. But it does mean the finite-sample constants cannot be instantiated on real data, and we prefer to say so rather than leave a reader to discover it. What does bind at our scale is the parameter bound $K^ = 1$, which trained networks routinely violate, and the σ_τ output layer. Those are the constraints our theory-versus-practice experiment enforces (Section 5); the value of the theory here lies in the ordering it predicts — which sample size limits accuracy, how n' should scale, which reference densities are admissible — rather than in numerical thresholds it can supply for a given dataset.*

Next, we conduct experiments with real-world data to evaluate the practical efficacy of synthetic

anomalies.

5 Experiments

5.1 Set-Up

We evaluate the area under the precision-recall curve (AUPR) of neural networks with vanilla classification (VC) [17]. We also test other AD methods (mentioned in order of their proximity in modeling VC): ES (VC with modified activation function) [27, 26], DROCC (VC but with adversarial synthetic anomalies) [15], ABC (VC with autoencoder structure) [54] and DeepSAD [40] (autoencoder with latent hypersphere classification). All methods are evaluated with and without our proposed method of including randomly sampled synthetic anomalies (SA). VC-SA models our theoretical framework. Here, we are interested in two research questions (RQs).

RQ1. Do synthetic anomalies improve performance of VC (i.e., VC-SA versus VC)?

RQ2. [Generalizability] Do synthetic anomalies improve performance of other state-of-the-art methods?

Results for RQ1 and RQ2 are reported in Tables 3a and 3b respectively. Both are reported in AUPR; Table 4 reports AUPR and AUROC side by side for VC and VC-SA on all five datasets. To avoid diluting the known anomaly supervision signal and contaminating the normal data during training, we *avoid adding too many synthetic anomalies*. Based on Theorem 4.11, we add $n' = n + n^-$ synthetic anomalies. More details are in Appendix G.

As a note, we also evaluate nine methods, each composing a binary classifier on an unsupervised AD method. These methods first do unsupervised AD to identify data that belong to the training classes, and then binary classifiers differentiate normal from known anomalous data given that the data are known. However, they consistently produce random (i.e., poor) performance and are unsuitable. We defer further details and discussions to Appendix G.4.

Dataset We summarize our five diverse real-world evaluation datasets spanning across **tabular**, **image** and **language** benchmarks. More details are in Appendix G.2. Our tabular datasets comprise NSL-KDD (cybersecurity) [49], Thyroid (medical) [37] and Arrhythmia (medical) [16]. MVTec [3] and AdvBench [9] are our image and language AD datasets respectively. Here, anomalies arise naturally from cyber-attacks, medical sickness, manufacturing defects and harmful text. For all datasets, we train with normal data and one type of “known” anomaly, and evaluate on normal data and the remaining anomalies in the dataset (mostly unknown anomalies). For instance, NSL-KDD has benign (normal) network traffic and 4 types of attacks (anomalies) during training and testing: Denial of Service (DoS), probe, remote access (RA), and privilege escalation (PE). To simulate semi-supervised AD, we use PE as known anomalies and the other 3 as unknown anomalies. To convert image and text data to tabular form, we use 1024-dimensional DINOv2 embeddings [34] and 384-dimensional BERT sentence embeddings [38] respectively.

In total, we have 24 unknown categories and 7 known categories. Due to the small dataset size of Arrhythmia and MVTec, known anomalies are used only in training and not testing, and all unknown anomaly types are grouped together as a large unknown anomaly class for evaluation. We emphasize more on unknown anomaly evaluation because unknowns characterize the AD problem more than knowns (see the common density level set estimation formulation in Section 3). Nevertheless, we also include known categories to gauge if synthetic anomalies will dilute known anomaly training signal, or if they can improve known anomaly performance.

Implementation We state the three details needed to reproduce our models, rather than deferring them to the appendix. *Architecture.* All our classifiers are feedforward multilayer perceptrons implemented in PyTorch, with constant hidden width and leaky ReLU activations, trained with the logistic loss. The theory is stated for plain ReLU under the hinge loss; we substitute both to avoid vanishing gradients and zero-loss samples, and we quantify what the substitution costs rather than assuming it away (Table ??, whose “theory class” column restores them). For NSL-KDD we use width $w = 678$; the remaining widths are 200 (Thyroid), 500 (Arrhythmia), 6000 (MVTec) and 6000 (AdvBench), chosen from the input dimension together with the sample size as Appendix G.5 describes. Depth is the one hyperparameter we tune: every reported model uses *two* hidden layers, which Appendix G.5 explains is far below the theory’s $L^* \geq 14$. Depth is counted in hidden layers throughout, as in Definition 3.2; two hidden layers means three weight matrices. The theoretical class of Definition 4.9 additionally imposes the sparsity budget v^* , the parameter bound K^* , and the σ_τ output layer; our default networks do *not* enforce these either, and Table ?? measures the whole gap at once. *Features.* Tabular data (NSL-KDD, Thyroid, Arrhythmia) are used directly, with numerical variables min-max normalized to $[0, 1]$ and categorical variables one-hot encoded. Image and text data are first mapped into a tabular space by frozen pre-trained encoders: 1024-dimensional DINOv2 embeddings [34] for MVTec and 384-dimensional BERT sentence embeddings [38] for AdvBench. No encoder is fine-tuned. *Support of $\mathcal{X}$.* We take $\mathcal{X}$ to be the smallest hyper-rectangle containing all normal training data, and synthetic anomalies are drawn inside it. A test point falling outside this box can be declared anomalous immediately; this case does not arise in our experiments, so it never affects the reported numbers.

5.2 Discussion

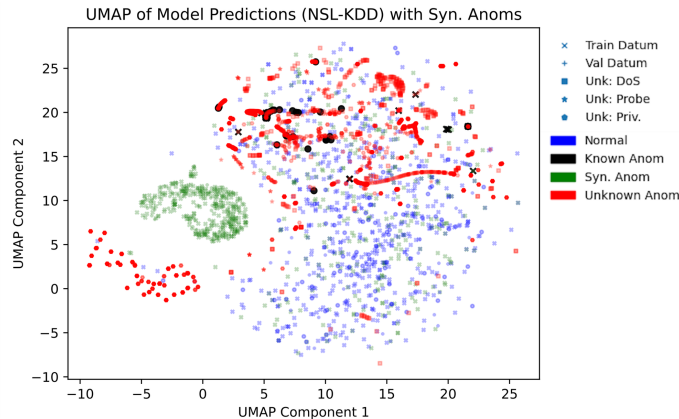


Figure 3: **Visualization.** Synthetic anomalies occupy regions with unknown anomalies (top right), training the model to classify unknown anomalies as anomalous.

RQ1. Is VC-SA better than VC? Across all 5 datasets, VC-SA generally outperforms VC. VC-SA expectedly has better performance on unknown anomalies (ahead of or level with VC on 19/24 unknown anomaly categories, strictly ahead on 18), with synthetic anomalies providing supervision signal to classify unknown regions as anomalous (Figure 3). Interestingly, VC-SA outperforms VC on known anomalies on 5/7 of known anomaly categories (subnormal from Thyroid and 4 of the 5 categories from AdvBench). Adding synthetic anomalies improves our modeling of density level set estimation (Case 1 in Figure 2), so improving (unsupervised) AD is not necessarily negatively

Table 3: **AUPR $\uparrow$ results with and without synthetic anomalies for (a) our theoretical model (vanilla classification, VC) and (b) other AD models.** Table 3b is a continuation of Table 3a, but separated as a different subtable to highlight the different RQs they are answering. Other models are arranged from left to right in order of how close they are to VC. More often than not, **synthetic anomalies (-SA suffix) generally improve results for VC and other AD models closer to the left (ES and DROCC).** Performance gains are seen for both unknown (unk.) and known anomalies. Meanwhile, autoencoder-based methods ABC and DeepSAD have more mixed results.

(a) RQ1. AUPR $\uparrow$ for our VC model.						(b) RQ2. AUPR $\uparrow$ for other methods. The Dataset and Anom. columns repeat those of Table 3a, so every row here is readable on its own rather than by positional alignment with the subtable beside it.									
Dataset	Type	Anom.	Random	VC	VC-SA (Ours)	Dataset	Anom.	ES	ES-SA	DROCC	DROCC-SA	ABC	ABC-SA	DeepSAD	DeepSAD-SA
NSL-KDD	Unk.	DoS	0.431	0.345 $\pm$ 0.036	0.793$\pm$0.055	NSL-KDD	DoS	0.716 $\pm$ 0.083	0.794$\pm$0.005	0.930$\pm$0.015	0.856 $\pm$ 0.067	0.935 $\pm$ 0.002	0.940$\pm$0.013	0.899$\pm$0.001	0.842 $\pm$ 0.001
		Probe	0.197	0.180 $\pm$ 0.010	0.649$\pm$0.078		Probe	0.423 $\pm$ 0.030	0.690$\pm$0.053	0.577 $\pm$ 0.037	0.634$\pm$0.205	0.900$\pm$0.018	0.860 $\pm$ 0.006	0.871$\pm$0.001	0.823 $\pm$ 0.003
		RA	0.218	0.543 $\pm$ 0.019	0.609$\pm$0.050		RA	0.565 $\pm$ 0.036	0.583$\pm$0.035	0.663$\pm$0.081	0.582 $\pm$ 0.062	0.546$\pm$0.034	0.507 $\pm$ 0.010	0.391 $\pm$ 0.001	0.520$\pm$0.003
Thyroid	Known	PE	0.007	0.609$\pm$0.001	0.486 $\pm$ 0.146	Thyroid	PE	0.606 $\pm$ 0.021	0.618$\pm$0.027	0.201$\pm$0.004	0.168 $\pm$ 0.036	0.226$\pm$0.065	0.054 $\pm$ 0.013	0.058$\pm$0.003	0.018 $\pm$ 0.000
	Unk.	Hyper.	0.023	0.565 $\pm$ 0.465	0.817$\pm$0.039		Hyper.	0.501 $\pm$ 0.006	0.588$\pm$0.032	0.064 $\pm$ 0.056	0.115$\pm$0.018	0.092 $\pm$ 0.001	0.154$\pm$0.003	0.221$\pm$0.004	0.161 $\pm$ 0.005
Arrhyth.	Known	Sub.	0.053	0.512 $\pm$ 0.380	0.751$\pm$0.020	Arrhyth.	Sub.	0.823 $\pm$ 0.027	0.826$\pm$0.027	0.051$\pm$0.004	0.045 $\pm$ 0.002	0.034 $\pm$ 0.000	0.039$\pm$0.000	0.053$\pm$0.000	0.049 $\pm$ 0.000
	Unk.	All	0.751	0.854$\pm$0.030	0.846 $\pm$ 0.003		All	0.826 $\pm$ 0.017	0.853$\pm$0.003	0.815 $\pm$ 0.016	0.842$\pm$0.005	0.896$\pm$0.002	0.890 $\pm$ 0.004	0.863 $\pm$ 0.000	0.873$\pm$0.001
MVTec (Image)	Unk.	Bottle	0.683	0.996 $\pm$ 0.001	0.997$\pm$0.000	MVTec (Image)	Bottle	0.999$\pm$0.000	0.999$\pm$0.000	0.999$\pm$0.003	0.999$\pm$0.000	0.987 $\pm$ 0.009	0.994$\pm$0.005	0.973$\pm$0.001	0.954 $\pm$ 0.002
		Cable	0.577	0.795 $\pm$ 0.013	0.868$\pm$0.005		Cable	0.816 $\pm$ 0.005	0.852$\pm$0.008	0.785 $\pm$ 0.006	0.856$\pm$0.009	0.843 $\pm$ 0.002	0.917$\pm$0.002	0.799 $\pm$ 0.003	0.802$\pm$0.006
		Capsule	0.789	0.908 $\pm$ 0.005	0.947$\pm$0.002		Capsule	0.911 $\pm$ 0.002	0.949$\pm$0.002	0.907 $\pm$ 0.005	0.956$\pm$0.002	0.964 $\pm$ 0.009	0.976$\pm$0.001	0.933$\pm$0.000	0.925 $\pm$ 0.001
		Carpet	0.714	0.998 $\pm$ 0.000	0.999$\pm$0.000		Carpet	0.908 $\pm$ 0.000	0.999$\pm$0.000	0.999 $\pm$ 0.001	1.000$\pm$0.000	0.994 $\pm$ 0.001	0.995$\pm$0.000	0.996$\pm$0.000	0.983 $\pm$ 0.000
		Carpent	0.789	0.998 $\pm$ 0.000	0.999$\pm$0.000		Carpent	0.905 $\pm$ 0.004	0.998$\pm$0.001	0.959 $\pm$ 0.014	0.997$\pm$0.000	0.992 $\pm$ 0.011	1.000$\pm$0.000	0.965$\pm$0.001	0.968$\pm$0.003
		Grid	0.682	0.999 $\pm$ 0.000	1.000$\pm$0.000		Grid	0.935 $\pm$ 0.036	0.958$\pm$0.005	0.911 $\pm$ 0.059	0.941$\pm$0.003	0.934 $\pm$ 0.063	0.952$\pm$0.002	0.810$\pm$0.002	0.792 $\pm$ 0.005
		Hazelnut	0.565	0.954$\pm$0.004	0.943 $\pm$ 0.009		Hazelnut	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	0.977 $\pm$ 0.006	1.000$\pm$0.000	0.998$\pm$0.000	0.998$\pm$0.001
		Leather	0.695	1.000$\pm$0.000	1.000$\pm$0.000		Leather	0.988$\pm$0.002	0.974 $\pm$ 0.010	0.983$\pm$0.001	0.973 $\pm$ 0.006	0.969 $\pm$ 0.002	0.974$\pm$0.001	0.943$\pm$0.001	0.941 $\pm$ 0.001
		Metal Nut	0.756	0.982$\pm$0.001	0.975 $\pm$ 0.006		Metal Nut	0.918$\pm$0.007	0.915 $\pm$ 0.043	0.898 $\pm$ 0.035	0.948$\pm$0.003	0.968 $\pm$ 0.002	0.971$\pm$0.002	0.944 $\pm$ 0.001	0.947$\pm$0.000
		Pill	0.817	0.925 $\pm$ 0.002	0.950$\pm$0.009		Pill	0.844$\pm$0.014	0.797 $\pm$ 0.102	0.830 $\pm$ 0.005	0.867$\pm$0.008	0.754 $\pm$ 0.027	0.933$\pm$0.005	0.709 $\pm$ 0.002	0.725$\pm$0.003
		Screw	0.699	0.839 $\pm$ 0.044	0.844$\pm$0.033		Screw	0.997 $\pm$ 0.000	0.998$\pm$0.000	0.997 $\pm$ 0.000	0.998$\pm$0.000	0.996 $\pm$ 0.000	0.998$\pm$0.000	0.991$\pm$0.000	0.989 $\pm$ 0.000
		Tile	0.670	0.980 $\pm$ 0.007	0.997$\pm$0.000		Tile	0.780 $\pm$ 0.019	0.815$\pm$0.066	0.795 $\pm$ 0.014	0.884$\pm$0.012	0.851 $\pm$ 0.017	0.900$\pm$0.008	0.836 $\pm$ 0.000	0.840$\pm$0.003
		Transistor	0.333	0.786 $\pm$ 0.026	0.872$\pm$0.010		Transistor	0.986 $\pm$ 0.001	0.995$\pm$0.002	0.992 $\pm$ 0.002	0.996$\pm$0.002	0.980 $\pm$ 0.001	0.997$\pm$0.001	0.979$\pm$0.001	0.977 $\pm$ 0.001
		Wood	0.732	0.984 $\pm$ 0.008	0.991$\pm$0.008		Wood	0.995$\pm$0.002	0.995$\pm$0.003	0.998$\pm$0.001	0.998$\pm$0.001	0.994 $\pm$ 0.001	0.999$\pm$0.000	0.990$\pm$0.000	0.988 $\pm$ 0.000
		Zipper	0.758	0.995 $\pm$ 0.002	0.998$\pm$0.001		Zipper	0.995$\pm$0.002	0.995$\pm$0.003	0.998$\pm$0.001	0.998$\pm$0.001	0.994 $\pm$ 0.001	0.999$\pm$0.000	0.990$\pm$0.000	0.988 $\pm$ 0.000
Adv-Bench (Text)	Unk.	satnews	0.082	0.798$\pm$0.028	0.232 $\pm$ 0.030	Adv-Bench (Text)	satnews	0.585$\pm$0.037	0.540 $\pm$ 0.023	0.283 $\pm$ 0.010	0.367$\pm$0.031	0.151$\pm$0.031	0.082 $\pm$ 0.003	0.080$\pm$0.000	0.074 $\pm$ 0.000
		CGFake	0.130	0.097$\pm$0.004	0.080 $\pm$ 0.002		CGFake	0.275$\pm$0.031	0.229 $\pm$ 0.016	0.150 $\pm$ 0.012	0.163$\pm$0.011	0.104$\pm$0.007	0.091 $\pm$ 0.004	0.090$\pm$0.000	0.084 $\pm$ 0.001
		jigsaw	0.130	0.185 $\pm$ 0.040	0.340$\pm$0.011		jigsaw	0.845$\pm$0.017	0.825 $\pm$ 0.030	0.754$\pm$0.028	0.715 $\pm$ 0.055	0.668$\pm$0.086	0.532 $\pm$ 0.060	0.107 $\pm$ 0.001	0.169$\pm$0.002
		EDENCE	0.113	0.102 $\pm$ 0.008	0.721$\pm$0.082		EDENCE	0.080$\pm$0.001	0.079 $\pm$ 0.001	0.101 $\pm$ 0.004	0.109$\pm$0.005	0.149 $\pm$ 0.018	0.156$\pm$0.010	0.159$\pm$0.001	0.137 $\pm$ 0.001
		FAS	0.140	0.087$\pm$0.002	0.126$\pm$0.007		FAS	0.737 $\pm$ 0.010	0.756$\pm$0.021	0.619 $\pm$ 0.044	0.712$\pm$0.014	0.555$\pm$0.127	0.090 $\pm$ 0.006	0.032$\pm$0.000	0.025 $\pm$ 0.000
	Known	LUN	0.074	0.762$\pm$0.032	0.532 $\pm$ 0.027		LUN	0.356 $\pm$ 0.009	0.375$\pm$0.014	0.335 $\pm$ 0.022	0.406$\pm$0.019	0.410$\pm$0.056	0.319 $\pm$ 0.011	0.144$\pm$0.002	0.114 $\pm$ 0.000
		amazon_lb	0.107	0.123 $\pm$ 0.017	0.824$\pm$0.036		amazon_lb	0.427 $\pm$ 0.054	0.444$\pm$0.070	0.154 $\pm$ 0.008	0.311$\pm$0.085	0.078$\pm$0.032	0.026 $\pm$ 0.001	0.023$\pm$0.001	0.021 $\pm$ 0.000
		HSOL	0.030	0.042 $\pm$ 0.009	0.731$\pm$0.015		HSOL	0.360 $\pm$ 0.027	0.387$\pm$0.039	0.195 $\pm$ 0.012	0.308$\pm$0.078	0.163$\pm$0.031	0.121 $\pm$ 0.009	0.076 $\pm$ 0.000	0.079$\pm$0.000
		assassin	0.022	0.048 $\pm$ 0.003	0.533$\pm$0.062		assassin	0.714 $\pm$ 0.029	0.760$\pm$0.026	0.594 $\pm$ 0.011	0.645$\pm$0.038	0.319$\pm$0.176	0.303 $\pm$ 0.044	0.108$\pm$0.000	0.102 $\pm$ 0.000
		enron	0.080	0.211 $\pm$ 0.037	0.361$\pm$0.015		enron	0.122 $\pm$ 0.003	0.138$\pm$0.005	0.119 $\pm$ 0.002	0.125$\pm$0.006	0.097 $\pm$ 0.004	0.101$\pm$0.002	0.131 $\pm$ 0.000	0.137$\pm$0.001

correlated with improving known anomaly performance, as seen here. Overall, **VC-SA performs better than VC.**

Both metrics, and why we report both. Table 4 reports AUPR and AUROC together. We lead with AUPR because the anomaly classes here are small — the random baseline is below 0.1 on eight of the 31 categories — and AUROC is optimistic under that imbalance [41]. That is a reason to read AUPR first, not a reason to withhold AUROC, so both are given for every dataset and category. The two agree on the RQ1 conclusion: VC-SA is ahead of or level with VC on 19 of 24 unknown-anomaly categories by AUPR and on 20 of 24 by AUROC. The one place they could be thought to disagree is not a metric artifact: on NSL-KDD’s DoS and Probe, VC is below the random baseline in AUPR *and* below 0.5 in AUROC, so both metrics say the same thing about it.

RQ2. How beneficial is adding random synthetic anomalies? Across datasets, synthetic anomalies produced the most performance gains in MVTec image AD regardless of method. This dataset has the highest dimension and fewest training samples. Here, known anomalies are the least dense, suggesting that the added synthetic anomalies increased the anomaly signal for improved performance.

Of other methods, ES and DROCC are the closest to VC and, likewise, benefit from adding synthetic anomalies. ES-SA outperforms ES in 14/24 (and tied 3) unknown and 7/7 known anomaly

Table 4: **AUPR** and **AUROC** results for our VC model with and without synthetic anomalies (-SA suffix) across five different datasets.

				AUPR		AUROC	
Dataset	Type	Anom.	Random	VC	VC-SA (Ours)	VC	VC-SA (Ours)
NSL-KDD	Unk.	DoS	0.431	0.345±0.036	0.793±0.055	0.331±0.131	0.807±0.057
		Probe	0.197	0.180±0.010	0.649±0.078	0.388±0.096	0.868±0.034
		RA	0.218	0.543±0.019	0.609±0.050	0.719±0.026	0.810±0.074
	Known	PE	0.007	0.609±0.001	0.486±0.146	0.986±0.006	0.988±0.001
Thyroid	Unk.	Hyper.	0.023	0.565±0.465	0.817±0.039	0.822±0.201	0.964±0.011
	Known	Sub.	0.053	0.512±0.380	0.751±0.020	0.856±0.192	0.985±0.003
Arrhyth.	Unk.	All	0.751	0.854±0.030	0.846±0.003	0.630±0.067	0.627±0.010
MVTec (Image)	Unk.	Bottle	0.683	0.996±0.001	0.997±0.000	0.991±0.002	0.994±0.001
		Cable	0.577	0.795±0.013	0.868±0.005	0.733±0.019	0.807±0.007
		Capsule	0.789	0.908±0.005	0.947±0.002	0.753±0.012	0.839±0.003
		Carpet	0.714	0.998±0.000	0.999±0.000	0.996±0.001	0.998±0.000
		Grid	0.682	0.999±0.000	1.000±0.000	0.998±0.000	0.999±0.000
		Hazelnut	0.565	0.954±0.004	0.943±0.009	0.919±0.008	0.920±0.013
		Leather	0.695	1.000±0.000	1.000±0.000	1.000±0.000	1.000±0.000
		Metal Nut	0.756	0.982±0.001	0.975±0.006	0.938±0.004	0.916±0.017
		Pill	0.817	0.925±0.002	0.950±0.009	0.713±0.001	0.801±0.025
		Screw	0.699	0.839±0.044	0.844±0.033	0.696±0.064	0.708±0.051
		Tile	0.670	0.980±0.007	0.997±0.000	0.970±0.010	0.994±0.001
		Transistor	0.333	0.786±0.026	0.872±0.010	0.826±0.007	0.875±0.009
		Wood	0.732	0.984±0.008	0.991±0.008	0.957±0.015	0.974±0.024
		Zipper	0.758	0.995±0.002	0.998±0.001	0.985±0.005	0.993±0.004
Adv-Bench (Text)	Unk.	satnews	0.082	0.798±0.028	0.232±0.030	0.977±0.004	0.793±0.029
		CGFake	0.130	0.097±0.004	0.080±0.002	0.390±0.027	0.242±0.019
		jigsaw	0.130	0.185±0.040	0.340±0.011	0.637±0.044	0.697±0.008
		EDENCE	0.113	0.102±0.008	0.721±0.082	0.473±0.029	0.941±0.017
		FAS	0.140	0.087±0.002	0.126±0.007	0.247±0.015	0.494±0.032
	Known	LUN	0.074	0.762±0.032	0.532±0.027	0.972±0.004	0.918±0.010
		amazon_lb	0.107	0.123±0.017	0.824±0.036	0.612±0.052	0.968±0.006
		HSOL	0.030	0.042±0.009	0.731±0.015	0.652±0.044	0.961±0.003
		assassin	0.022	0.048±0.003	0.533±0.062	0.761±0.016	0.931±0.002
		enron	0.080	0.211±0.037	0.361±0.015	0.765±0.010	0.820±0.013

categories, while DROCC-SA outperforms DROCC in 17/24 (and tied 3) unknown and 5/7 known anomaly categories. Consistent performance gains demonstrate that adding **synthetic anomalies generalizes well to other classification-based AD methods**.

Meanwhile, ABC and DeepSAD enforce autoencoder (i.e., encoder-decoder) structures. ABC is the next closest to VC, using a binary classification objective with an autoencoder structure. ABC-SA outperforms ABC in 17/24 unknown anomaly categories, but most (14) improvements come from one dataset (MVTec); performance on other datasets is mixed. DeepSAD is the least similar to VC with a two-stage training procedure: first training an autoencoder, then using the encoder for binary classification. DeepSAD-SA outperforms DeepSAD in only 8/24 (and tied 1) unknown anomaly categories and is the only model where adding synthetic anomalies is not better. Notably, DeepSAD has good performance on DoS and Probe anomalies in NSL-KDD, which are easy anomalies [28], but struggles on other anomalies (e.g., Thyroid and AdvBench). Here, DeepSAD already underperforms, and adding synthetic anomalies may not be the solution for that. Moreover, only 2/7 known anomaly categories are better for both ABC-SA vs. ABC and DeepSAD-SA vs. DeepSAD, suggesting that synthetic anomalies dilute known anomaly training signal for these autoencoder models.

Comparison with Unsupervised AD Because our formulation reduces to unsupervised AD when the reference density is constant, it is fair to ask what the known anomaly sample actually buys. We therefore run three unsupervised detectors — OCSVM, Isolation Forest and DeepSVDD — on all five datasets, trained on normal data alone. The pattern is the one the reference-density account predicts: semi-supervised AD is stronger on known anomaly categories, where the labeled sample is informative, while unsupervised AD is more even across unknown categories, where it has no misleading prior to overcome. Adding synthetic anomalies moves our classifier toward the second behavior without giving up the first, which is precisely the trade the mixture $h_2 = \tilde{s}h_- + (1 - \tilde{s})$ is designed to make.

We deliberately do *not* compare against specialised unsupervised vision detectors such as PatchCore, EfficientAD or CFlow. The reason is concrete rather than a matter of scope: those methods operate on *spatial patch features* extracted from raw images and score anomalies by local patch deviation, whereas every method in this paper — ours and all baselines — consumes a single pooled embedding per input and is modality-agnostic by construction. Benchmarking them together on MVTec would compare a general framework against detectors engineered for one modality, and would report mostly the value of patch-level locality rather than anything about the reference density, which is the question at issue. A meaningful comparison would require re-deriving our framework at the patch level, which we regard as a worthwhile extension rather than a control experiment.

Limitations and Extensions Overall, we validate that adding synthetic anomalies to classification-based AD methods works well, while it generally works less effectively with autoencoder constraints. Supervision for negatives (i.e., anomalies) must be more carefully designed in autoencoders (e.g., in contrastive learning), but we leave this to future work. Additionally, there could be other valid semi-supervised AD formulations other than ours.

Ablations Due to space constraints, we leave details in Appendix G.5, summarizing ablations across three key hyperparameters: width, depth and number of synthetic anomalies (Table 5). Wider and deeper networks provide higher expressivity, but we observe vanishing gradients in the latter. Performance is not as sensitive to width. Meanwhile, more synthetic anomalies (even a small amount) improves unknown anomaly performance, but contaminates the supervision signal from known anomalies, hence affecting known anomaly performance. Therefore, in our experiments, we choose

Table 5: **Ablations for NSL-KDD** for the width w , depth L (number of hidden layers, as in Definition 3.2) and proportion of synthetic anomalies n' to real training data $r := n + n^-$. AUPR $\uparrow$ of our vanilla classifier reported across attacks (anomalies).

Model\Attack	Unknown Anomalies			Known PE
	DoS	Probe	RA	
Random	0.431	0.197	0.218	0.007
$w = 300$	0.756 $\pm$ 0.048	0.658 $\pm$ 0.106	0.584 $\pm$ 0.025	0.309 $\pm$ 0.026
$w = 678$ (Ours)	0.793 $\pm$ 0.055	0.649 $\pm$ 0.078	0.609 $\pm$ 0.050	0.486 $\pm$ 0.146
$w = 1500$	0.772 $\pm$ 0.034	0.547 $\pm$ 0.068	0.602 $\pm$ 0.025	0.486 $\pm$ 0.077
$L = 2$ (Ours)	0.793 $\pm$ 0.055	0.649 $\pm$ 0.078	0.609 $\pm$ 0.050	0.486 $\pm$ 0.146
$L = 3$	0.720 $\pm$ 0.023	0.337 $\pm$ 0.063	0.581 $\pm$ 0.035	0.560 $\pm$ 0.116
$L = 8$	0.431 $\pm$ 0.000	0.197 $\pm$ 0.000	0.218 $\pm$ 0.000	0.007 $\pm$ 0.000
$L = 17$	0.431 $\pm$ 0.000	0.197 $\pm$ 0.000	0.218 $\pm$ 0.000	0.007 $\pm$ 0.000
$n' = 0r$	0.345 $\pm$ 0.036	0.180 $\pm$ 0.010	0.543 $\pm$ 0.019	0.609 $\pm$ 0.001
$n' = 0.001r$	0.744 $\pm$ 0.009	0.651 $\pm$ 0.049	0.598 $\pm$ 0.030	0.513 $\pm$ 0.082
$n' = r$ (Ours)	0.793 $\pm$ 0.055	0.649 $\pm$ 0.078	0.609 $\pm$ 0.050	0.486 $\pm$ 0.146
$n' = 5r$	0.801 $\pm$ 0.026	0.649 $\pm$ 0.070	0.633 $\pm$ 0.063	0.431 $\pm$ 0.158
$n' = 20r$	0.763 $\pm$ 0.051	0.503 $\pm$ 0.152	0.553 $\pm$ 0.087	0.298 $\pm$ 0.070

depth and width to balance expressivity (determined by data dimension and number of samples), and $n' = n + n^-$.

6 Conclusion

Semi-supervised AD offers a way to incorporate known anomalies, but its connection to unsupervised AD has remained unclear. In this work, we establish this connection by placing both settings on one axis: the choice of reference density in a likelihood-ratio test. Unsupervised AD sits at a constant reference, binary classification at the known-anomaly density, and semi-supervised AD at the mixture that synthetic anomalies realize. This choice enables (i) anomaly modeling in low-density regions and (ii) an explicit convergence rate for ReLU network classifiers trained with a non-representative labeled anomaly sample. Experiments on five diverse benchmarks reveal consistent performance gains for our theoretical model *and* other classification-based AD methods. These theoretical and empirical results highlight that using synthetic anomalies in AD is generalizable. We envision future work that extends our theoretical insights to other AD settings (e.g., when contamination is present in normal data).

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Appendix

A Discussions on the Bayes Classifier, and the Bayes Rule minimizing Generalization Error

Recall that $s \in (0, 1)$ represents the proportion of normal data on $\mathcal{X}$. For a pair of random variable $(X, Y) \in \mathcal{X} \times \mathcal{Y}$, we define $P(Y = 1) = s$ and $P(Y = -1) = 1 - s$. For any function f on $\mathcal{X} \times \mathcal{Y}$, we have

$$\int_{\mathcal{X} \times \mathcal{Y}} f(X, Y) dP = s \int_{\mathcal{X}} f(X, 1) h_1 d\mu + (1 - s) \int_{\mathcal{X}} f(X, -1) h_2 d\mu. \quad (\text{A.1})$$

From (A.1), we derive the conditional class probability function as

$$\eta(X) := P(Y = 1|X) = \frac{s \cdot h_1(X)}{s \cdot h_1(X) + (1 - s) \cdot h_2(X)}. \quad (\text{A.2})$$

We can also derive the regression function, denoted by f_P , given as

$$\begin{aligned} f_P(X) &:= E[Y|X] \\ &= 1 \cdot P(Y = 1|X) + (-1) \cdot P(Y = -1|X) \\ &= \frac{s \cdot h_1(X) - (1 - s) \cdot h_2(X)}{s \cdot h_1(X) + (1 - s) \cdot h_2(X)}, \quad \forall X \in \mathcal{X}. \end{aligned}$$

Both η and f_P are fundamental in characterizing the classification problem and assessing the performance of neural network classifiers.

Throughout this work, we adopt the **hinge loss**, defined as

$$\phi(u) := \max\{1 - u, 0\},$$

which is one of the most widely used convex loss functions in binary classification. While the 0-1 loss is the most natural choice for binary classification, minimizing the empirical risk w.r.t. it is computationally intractable (NP-hard) due to its non-convex and discontinuous nature [2]. To overcome this challenge, convex surrogate loss functions $V : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$ are often employed to make computation feasible. Among these, learning a neural network classifier with hinge loss is relatively straightforward owing to the gradient descent algorithm [33, 14].

The generalization error for $f : \mathcal{X} \rightarrow \mathbb{R}$ w.r.t. the hinge loss ϕ and the probability measure P is defined as

$$\begin{aligned} \varepsilon(f) &= \int_{\mathcal{X} \times \mathcal{Y}} \phi(Yf(X)) dP \\ &= \int_{\mathcal{X}} \int_{\mathcal{Y}} \phi(Yf(X)) dP(Y|X) dP_{\mathcal{X}} \\ &= \int_{\mathcal{X}} [\phi(f(X))P(Y = 1|X) + \phi(-f(X))P(Y = -1|X)] dP_{\mathcal{X}} \\ &= \int_{\mathcal{X}} [s\phi(f(X))h_1(X) + (1 - s)\phi(-f(X))h_2(X)] d\mu. \end{aligned}$$

The following proposition establishes that the Bayes classifier f_c also minimizes the generalization error. In fact, f_c can be defined explicitly to be

$$f_c(X) = \text{sign}(f_P(X)).$$

Its proof is given below.

Proposition A.15. *The generalization error $\varepsilon(f)$ is minimized by the Bayes rule; that is, for all $X \in \mathcal{X}$,*

$$\begin{aligned} f_c(X) &= \text{sign}(f_P(X)) \\ &= \text{sign}(s \cdot h_1(X) - (1-s) \cdot h_2(X)) \\ &= \begin{cases} 1, & \text{if } s \cdot h_1(X) - (1-s) \cdot h_2(X) \geq 0, \\ -1, & \text{if } s \cdot h_1(X) - (1-s) \cdot h_2(X) < 0. \end{cases} \end{aligned}$$

Proof of Proposition A.15. A minimizer f^* of $\varepsilon(f)$ can be found by taking its value $f^*(X)$ at every $X \in \mathcal{X}$ to be a minimum of the convex function $\Phi = \Phi_X$ defined by

$$\Phi(u) = s \cdot h_1(X)\phi(u) + (1-s) \cdot h_2(X)\phi(-u), \quad u \in \mathbb{R}. \quad (\text{A.3})$$

Notice that the hinge loss $\phi(u) = \max\{1-u, 0\}$ is not differentiable at $u = 1$. Thus, to find a minimizer of the function Φ , we need to look at its one-sided derivatives.

Let $\phi'_-(u)$ and $\phi'_+(u)$ denote the left derivative and right derivative of $\phi(u)$, respectively. The one-sided derivatives of the hinge loss $\phi(u)$ are

$$\phi'_-(u) = \begin{cases} -1, & \text{if } u \leq 1, \\ 0, & \text{if } u > 1, \end{cases} \quad \text{and} \quad \phi'_+(u) = \begin{cases} -1, & \text{if } u < 1, \\ 0, & \text{if } u \geq 1. \end{cases} \quad (\text{A.4})$$

Moreover, the left derivative of $\phi(-u)$ is

$$\begin{aligned} \phi'_-(-u) &= \lim_{x \rightarrow 0^-} \frac{\phi(-(u+x)) - \phi(-u)}{x} \\ &= \lim_{x \rightarrow 0^-} -\frac{\phi(-u-x) - \phi(-u)}{-x} \\ &= -\phi'_+(-u). \end{aligned}$$

It follows that the left derivative of the function Φ is given by

$$\begin{aligned} \Phi'_-(u) &= s \cdot h_1(X)\phi'_-(u) + (1-s) \cdot h_2(X)\phi'_-(-u) \\ &= s \cdot h_1(X)\phi'_-(u) - (1-s) \cdot h_2(X)\phi'_+(-u) \\ &= \begin{cases} (1-s) \cdot h_2(X), & \text{if } u > 1, \\ (1-s) \cdot h_2(X) - s \cdot h_1(X), & \text{if } -1 < u \leq 1, \\ -s \cdot h_1(X), & \text{if } u \leq -1. \end{cases} \end{aligned}$$

Similarly, the right derivative of the function Φ is given by

$$\begin{aligned} \Phi'_+(u) &= s \cdot h_1(X)\phi'_+(u) + (1-s) \cdot h_2(X)\phi'_+(-u) \\ &= s \cdot h_1(X)\phi'_+(u) - (1-s) \cdot h_2(X)\phi'_-(-u) \\ &= \begin{cases} (1-s) \cdot h_2(X), & \text{if } u \geq 1, \\ (1-s) \cdot h_2(X) - s \cdot h_1(X), & \text{if } -1 \leq u < 1, \\ -s \cdot h_1(X), & \text{if } u < -1. \end{cases} \end{aligned}$$

It follows that a minimizer u^* of Φ must be on $[-1, 1]$. Also, the convexity of Φ implies that a minimizer u^* must satisfies

$$\Phi'_-(u^*) = \begin{cases} (1-s) \cdot h_2(X) - s \cdot h_1(X) \leq 0, & \text{if } -1 < u^* \leq 1, \\ -s \cdot h_1(X) \leq 0, & \text{if } u^* = -1. \end{cases} \quad (\text{A.5})$$

and

$$\Phi'_+(u^*) = \begin{cases} (1-s) \cdot h_2(X) \geq 0, & \text{if } u^* = 1, \\ (1-s) \cdot h_2(X) - s \cdot h_1(X) \geq 0, & \text{if } -1 \leq u^* < 1. \end{cases} \quad (\text{A.6})$$

We consider all the possible cases:

- If $(1-s) \cdot h_2(X) - s \cdot h_1(X) < 0$, the minimizer u^* must be 1; this is because $(1-s) \cdot h_2(X) - s \cdot h_1(X) \geq 0$ for all $u^* \in [-1, 1)$.
- Similarly, if $(1-s) \cdot h_2(X) - s \cdot h_1(X) > 0$, the minimizer u^* must be -1 ; this is because $(1-s) \cdot h_2(X) - s \cdot h_1(X) \leq 0$ for all $u^* \in (-1, 1]$.
- If $(1-s) \cdot h_2(X) - s \cdot h_1(X) = 0$, the function Φ is just constant on $[-1, 1]$ so a minimizer u^* can be any number on $[-1, 1]$. We take $u^* = 1$.

This completes the proof. $\square$

B Remarks about Theoretical Analysis

B.1 Remarks about the Tsybakov noise condition

Here, we provide a detailed discussion and elaboration on the well-known Tsybakov noise condition, which is utilized in this work to model the classification problem.

Remark B.16. *Throughout this work, we assume the Tsybakov noise condition (see, e.g., [32, 51]), which is a common assumption imposed in the literature in classification. This condition describes the behavior of the regression function $f_P(X) = \mathbb{E}[Y|X]$ (with parameter q) around the decision boundary $\{x : f_P(x) = 0\}$. Specifically, it assumes that for some constant $c_0 > 0$ and $q \in [0, \infty)$,*

$$\mathbb{P}_X(\{X \in \mathcal{X} : |f_P(X)| \leq t\}) \leq c_0 t^q, \quad \forall t > 0, \quad (\text{B.1})$$

where q is commonly referred to as the noise exponent. Intuitively, bigger q means there is a higher chance that $f_P(X)$ is bounded away from 0, which is favorable for classification. If q approaches ∞ , the regression function f_P is bounded away from 0 for P_X -almost all $x \in \mathcal{X}$. Conversely, smaller q implies that there is a plateau behavior near the boundary 0, which is considered a difficult situation for classification. Note that the assumption becomes trivial for $q = 0$ because every probability measure satisfies (B.1) by taking the constant $c_0 = 1$.

Equivalently, the noise condition can be stated in terms of $\eta(X) = \mathbb{P}(Y = 1|X)$ as

$$\mathbb{P}_X(\{X \in \mathcal{X} : |2\eta(X) - 1| \leq t\}) \leq c_0 t^q. \quad (\text{B.2})$$

The statements (B.1) and (B.2) are equivalent because

$$f_P(X) = \eta(X) - (1 - \eta(X)) = 2\eta(X) - 1.$$

See [51, 1, 2] for further discussion on this assumption.

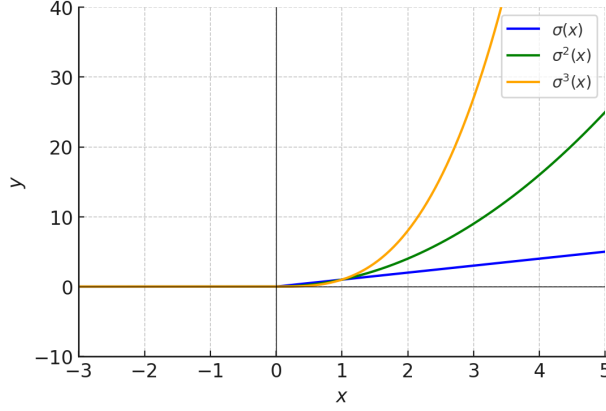


Figure 4: Plot of the ReLU function $\sigma(x) = \max(0, x)$ and its squared and cubed variants, $\sigma^2(x)$ and $\sigma^3(x)$, illustrating their increasing smoothness and growth rates for $x > 0$.

B.2 Remarks about the “approx-sign” function

Remark B.17. Recall that σ^k is the ReLU^k activation function with $k \in \mathbb{N}$, which is the generalization of ReLU defined as

$$\sigma^k(x) = (\max\{0, x\})^k.$$

These higher-order variants of the ReLU functions preserve the non-negativity and sparsity of ReLU while introducing smoother transitions near the origin. See Figure 4 for a plot comparing $\sigma(x)$, $\sigma^2(x)$, and $\sigma^3(x)$.

Recall the “approx-sign” function $\sigma_\tau : \mathbb{R} \rightarrow [-1, 1]$, defined previously in equation (3.7) with a bandwidth parameter $\tau > 0$, as

$$\sigma_\tau(x) = \frac{1}{\tau}\sigma(x) - \frac{1}{\tau}\sigma(x - \tau) - \frac{1}{\tau}\sigma(-x) + \frac{1}{\tau}\sigma(-x - \tau),$$

which simplifies to the piecewise form:

$$\sigma_\tau(x) = \begin{cases} 1, & \text{if } x \geq \tau, \\ \frac{x}{\tau}, & \text{if } x \in [-\tau, \tau), \\ -1, & \text{if } x < -\tau. \end{cases}$$

We see that this function is a linear combination of four scaled ReLU units. Moreover, it is a smoothed sign function that transitions linearly between -1 and 1 within the interval $[-\tau, \tau)$. Outside this interval, it behaves like the standard sign function. As $\tau \rightarrow 0$, $\sigma_\tau(x)$ converges to the sign function. We use the function σ_τ to approximate the sign function.

We also define higher-order approximations $\sigma_\tau^k(x)$, which are smoother versions of $\sigma_\tau(x)$, constructed using σ^k functions for $k \in \mathbb{N}$. Specifically, the function $\sigma_\tau^k(x)$ is defined as

$$\sigma_\tau^k(x) := \frac{1}{k!\tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(x - j\tau) - \frac{1}{k!\tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(-x - j\tau).$$

See Figure 5 for visual comparison using $\tau = 0.1$.

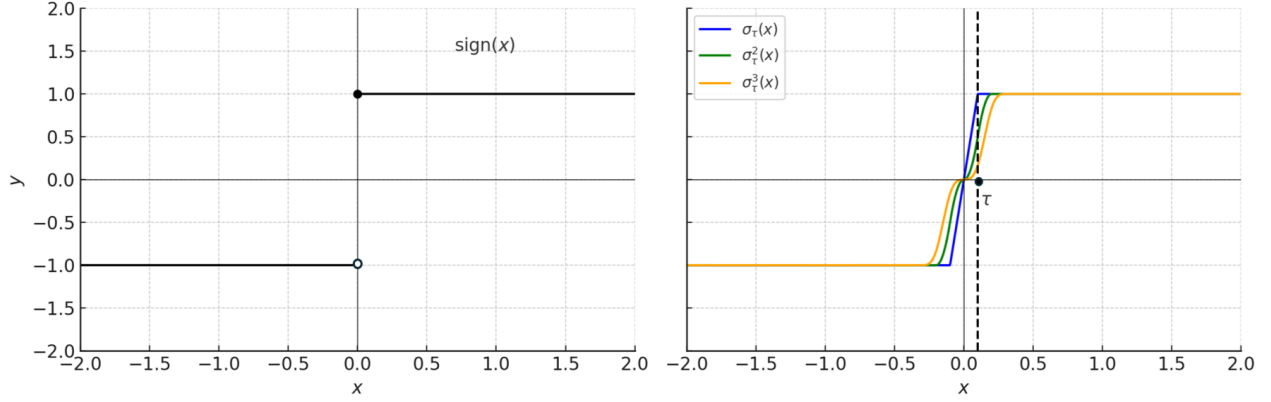


Figure 5: Comparison of the sign function and the “approx-sign” functions, which are its smoothed localized approximations. The left-hand plot shows the discontinuous sign function, defined as $\text{sign}(x) = -1$ for $x < 0$ and 1 for $x \geq 0$. The right-hand plot displays the “approx-sign” functions $\sigma_\tau(x)$, $\sigma_\tau^2(x)$, and $\sigma_\tau^3(x)$. They are constructed using combinations of scaled ReLU $\sigma(x)$ and their powers $\sigma^2(x)$, $\sigma^3(x)$ to create increasingly smooth and localized bump-like functions. In this example, we set $\tau = 0.1$.

B.3 Significance and implications of Theorem 3.3

Remark B.18. Here, we discuss and highlight the novelty of Theorem 3.3. In binary classification, the generalization error $\varepsilon(f)$ is minimized by the Bayes rule (as shown in Proposition A.15). The Bayes classifier $f_c = \text{sign}(f_P)$ is typically discontinuous.

Existing literature often estimates the excess generalization error $\varepsilon(f) - \varepsilon(f_c)$ by the L_1 norm (w.r.t. marginal distribution on $\mathcal{X}$) of the difference between f and f_c [8, 30]. However, due to the discontinuous nature of f_c , the L_1 error usually decays slowly to zero.

In contrast, Theorem 3.3 shows that as long as the error in approximating the regression function f_P (instead of f_c) converges fast to zero, the excess generalization error (and consequently the excess risk $R(f) - R(f_c)$) will also converge fast to zero.

Our results leverage Tsybakov’s noise condition to establish a relationship between the excess generalization error of f and the approximation error $\|f - f_P\|_{L^\infty}$. Since the regression function f_P is smooth (particularly when synthetic anomalies are incorporated during training), errors of approximating this smooth regression function can be small and decay fast, as supported by many existing results in approximation theory (e.g., [56, 45]). This approach offers an alternative estimate of the excess generalization error in classification, leveraging on both the noise condition and the smoothness of the regression function.

C Examples

In this section, we present two examples, each based on specific choices of the density functions h_1 and h_- , to illustrate two key points: (i) how the resulting regression function f_P can be discontinuous, and (ii) how the inclusion of synthetic anomalies can improve the smoothness of f_P .

Example C.19. Here, let us consider a simple example of h_1, h_- and the corresponding regression function.

Let H be the univariate hat function on $[-1, 1]$ given by

$$H(x) = \max\{1 - |x|, 0\},$$

where it peaks at $x = 0$ and linearly decreases to 0 at $x = -1$ and $x = 1$. We let

$$h_1(X) = 4H(4X - 3)$$

and

$$h_-(X) = 4H(4X - 1)$$

be two hat functions supported on $[1/2, 1]$ and $[0, 1/2]$ respectively. From Figure 6, we see that both functions are exactly zero outside their respective supports. Also, the intersection of $[0, 1/2]$ and $[1/2, 1]$ has an empty interior. This presents a specific example that corresponds to the general scenarios illustrated in Proposition 3.1.

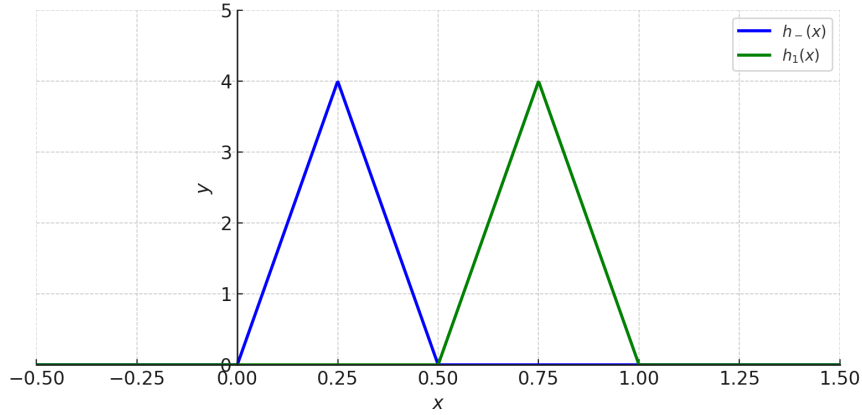


Figure 6: The plot showcases the density functions h_- and h_1 given in Example C.19, which are two hat functions supported on $[0, 1/2]$ and $[1/2, 1]$, respectively.

When $\tilde{s} = 1$, so that no synthetic anomalies are drawn and $h_2(X) = h_-(X)$, the regression function is given by

$$f_P(X) = \begin{cases} -1, & \text{if } X \in [0, 1/2], \\ 1, & \text{if } X \in (1/2, 1], \end{cases}$$

which is a function discontinuous on $[0, 1]$.

On the other hand, if we include synthetic anomalies in training and let $h_2(X) = \tilde{s}h_-(X) + (1 - \tilde{s})$ for some $\tilde{s} < 1$, then h_- vanishes on $(1/2, 1]$ and so $h_2 \equiv 1 - \tilde{s}$ there, and the regression function becomes

$$f_P(X) = \begin{cases} -1, & \text{if } X \in [0, 1/2], \\ \frac{16s(X-1/2)-(1-s)(1-\tilde{s})}{16s(X-1/2)+(1-s)(1-\tilde{s})}, & \text{if } X \in (1/2, 3/4], \\ \frac{16s(1-X)-(1-s)(1-\tilde{s})}{16s(1-X)+(1-s)(1-\tilde{s})}, & \text{if } X \in (3/4, 1], \end{cases}$$

which is continuous. In particular, observe that for $X \in (1/2, 3/4)$,

$$f'_P(X) = \frac{32s(1-s)(1-\tilde{s})}{(16s(X-1/2)+(1-s)(1-\tilde{s}))^2}$$

and the Lipschitz constant (C^1 seminorm) of f_P equals $\frac{32s}{(1-s)(1-\tilde{s})}$, attained as $X \downarrow 1/2$. We can see that the regularity of f_P gets better as $\tilde{s}$ becomes smaller, that is, as the proportion $1 - \tilde{s}$ of synthetic anomalies grows. Figure 7 provides a visual illustration of the resulting regression function for various choices of $\tilde{s} < 1$. We can see that the smaller the $\tilde{s}$, the smoother the f_P .

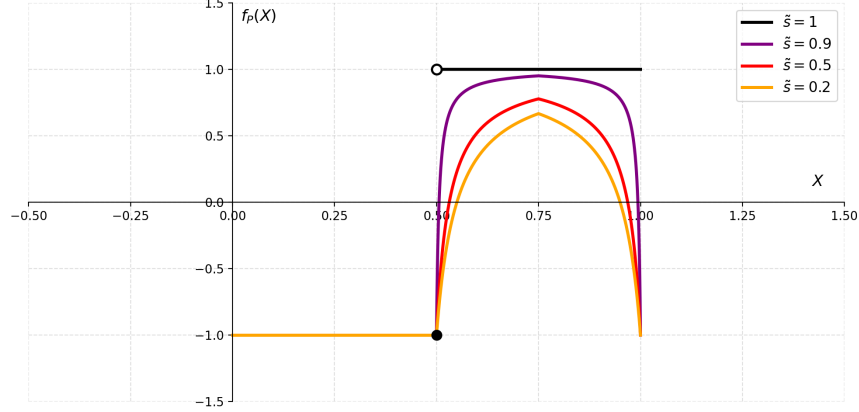


Figure 7: A comparison of the regression function $f_P(X)$ for varying parameter values $\tilde{s} = 1, \tilde{s} = 0.9, \tilde{s} = 0.5, \tilde{s} = 0.2$. The case $\tilde{s} = 1$ corresponds to the absence of synthetic anomalies and results in a discontinuous function with a jump from -1 to 1 . Decreasing the parameter $\tilde{s}$ (i.e., adding more synthetic anomalies, which enter with weight $1 - \tilde{s}$) results in progressively smoother functions. Here, we take $s = 0.5$.

The above example considers one-dimensional density functions. We now proceed to present an example involving d -dimensional density functions, demonstrating how the inclusion of synthetic anomalies can improve the smoothness of the resulting regression function. The example is given below.

Example C.20. Consider the univariate hat function defined on the interval $[-1/2, 1/2]$ given by

$$\max\{2 - 4|x|, 0\}, \quad (\text{C.1})$$

where it peaks at $x = 0$ and linearly decreases to 0 at $x = -1/2$ and $x = 1/2$. Now consider a d -dimensional function where each dimension is a hat function, that is,

$$H(X) = H(X_1, \dots, X_d) = \prod_{i=1}^d H(X_i) = \prod_{i=1}^d \max\{2 - 4|X_i|, 0\}. \quad (\text{C.2})$$

This function is non-zero only within the region $X_i \in [-1/2, 1/2]$ for all i , and decays continuously to zero as any $|X_i|$ approaches $1/2$. We can easily verify that this function is Lipschitz continuous, i.e. α -Hölder with $\alpha = 1$; it is not differentiable at the kinks.

Suppose the normal density function h_1 is given by

$$h_1(X) = H(X + (1/2, 0, 0, \dots, 0)), \quad (\text{C.3})$$

and suppose the real anomaly density function h_- is given by

$$h_-(X) = H(X - (1/2, 0, 0, \dots, 0)). \quad (\text{C.4})$$

If $h_2 = h_-$ without synthetic data, the regression function is

$$\begin{aligned} f_P(X) &= \frac{s \cdot h_1(X) - (1-s) \cdot h_-(X)}{s \cdot h_1(X) + (1-s) \cdot h_-(X)} \\ &= \begin{cases} \frac{s \cdot h_1(X)}{s \cdot h_1(X)} = 1, & \text{if } X \in [-1, 0] \times [-1/2, 1/2]^{d-1}, \\ -\frac{(1-s) \cdot h_-(X)}{(1-s) \cdot h_-(X)} = -1, & \text{if } X \in (0, 1] \times [-1/2, 1/2]^{d-1}. \end{cases} \end{aligned}$$

We can see that f_P is not continuous on the line segment $X_1 = 0$. With the choice $h_2 = h_-$, the regression function has lost its Lipschitz continuity. Learning a discontinuous function by neural network is generally difficult in theory. However, if we include synthetic anomalies and set $h_2 = \tilde{s}h_- + (1-\tilde{s})$ for $\tilde{s} < 1$, then the regression function becomes

$$f_P(X) = \frac{s \cdot h_1(X) - (1-s)\tilde{s} \cdot h_-(X) - (1-s)(1-\tilde{s})}{s \cdot h_1(X) + (1-s)\tilde{s} \cdot h_-(X) + (1-s)(1-\tilde{s})}$$

We can verify that this regression function is Lipschitz continuous. In fact, because we have the positive constant term $(1-s)(1-\tilde{s})$ in the denominator, it helps in maintaining the smoothness of f_P .

D Proofs of the Propositions and of (3.4)

This section proves (3.4) (via Lemma D.21), Proposition 3.1 and Proposition 4.5.

D.1 Proof of Equation (3.4)

We first present the following lemma that showcases how the Tsybakov noise condition relates to our data distribution on the domain $\mathcal{X}$. This lemma will be instrumental in proving the statement of Equation (3.4).

Lemma D.21. *Let μ be a known probability measure on $\mathcal{X}$. Let Q and W be distributions on $\mathcal{X}$ such that Q has a density h_1 with respect to μ , and W has a density h_2 with respect to μ . For $s \in (0, 1)$, the Tsybakov noise condition (3.3) with noise exponent $q \in [0, \infty)$ is satisfied if and only if*

$$\int_{\left\{X \in \mathcal{X}: \left| \frac{h_1(X)}{h_2(X)} - \frac{1-s}{s} \right| \leq \left(\frac{1-s}{s} \right) t\right\}} h_2(X) d\mu = \mathcal{O}(t^q). \quad (\text{D.1})$$

In other words, $\frac{h_1}{h_2}$ has $\frac{1-s}{s}$ -exponent q .

Proof of Lemma D.21. In our problem setting, we observe that the regression function depends on both densities h_1 and h_2 . For X with $h_2(X) > 0$, the regression function can be expressed as

$$f_P(X) = \frac{s \cdot \frac{h_1(X)}{h_2(X)} - (1-s)}{s \cdot \frac{h_1(X)}{h_2(X)} + (1-s)}.$$

Note that the case $h_2(X) = 0$ means $f_P(X) = 1$ and the point X is outside the domain stated in (3.3). It follows that for $t \in (0, 1/4)$, $|f_P(X)| \leq t$ if and only if

$$\frac{1-s}{s} \cdot \frac{1-t}{1+t} \leq \frac{h_1(X)}{h_2(X)} \leq \frac{1-s}{s} \cdot \frac{1+t}{1-t}, \quad (\text{D.2})$$

which is the same as

$$-\frac{1-s}{s} \cdot \frac{2t}{1+t} \leq \frac{h_1(X)}{h_2(X)} - \frac{1-s}{s} \leq \frac{1-s}{s} \cdot \frac{2t}{1-t}. \quad (\text{D.3})$$

Thus, for $t \in (0, 1/4)$, we have

$$\begin{aligned} P_X(\{X \in \mathcal{X} : |f_P(X)| \leq t\}) &= \int_{\{X \in \mathcal{X} : |f_P(X)| \leq t\}} dP_X \\ &= \int_{X_t} [s \cdot h_1(X) + (1-s) \cdot h_2(X)] d\mu \\ &= \int_{X_t} \left(s \cdot \frac{h_1(X)}{h_2(X)} + (1-s) \right) h_2(X) d\mu, \end{aligned}$$

where $X_t := \left\{ X \in \mathcal{X} : -\frac{1-s}{s} \cdot \frac{2t}{1+t} \leq \frac{h_1(X)}{h_2(X)} - \frac{1-s}{s} \leq \frac{1-s}{s} \cdot \frac{2t}{1-t} \right\}$.

Then, we can get an upper bound for the measure by observing

$$s \cdot \frac{h_1(X)}{h_2(X)} \leq (1-s) \frac{1+t}{1-t} \leq (1-s) \frac{1+1/4}{1-1/4} = \frac{5(1-s)}{3} \leq 2(1-s);$$

on X_t and enlarging the domain X_t as

$$\begin{aligned} P_X(\{X \in \mathcal{X} : |f_P(X)| \leq t\}) &= \int_{X_t} \left(s \cdot \frac{h_1(X)}{h_2(X)} + (1-s) \right) h_2(X) d\mu \\ &\leq 3(1-s) \int_{\left\{ X \in \mathcal{X} : \left| \frac{h_1(X)}{h_2(X)} - \frac{1-s}{s} \right| \leq \frac{1-s}{s} \cdot 3t \right\}} h_2(X) d\mu. \end{aligned}$$

A lower bound can be derived by observing

$$s \cdot \frac{h_1(X)}{h_2(X)} \geq (1-s) \frac{1-t}{1+t} \geq (1-s) \frac{1-1/4}{1+1/4} = \frac{3(1-s)}{5}$$

on X_t and reducing the domain X_t as

$$\begin{aligned} P_X(\{X \in \mathcal{X} : |f_P(X)| \leq t\}) &= \int_{X_t} \left(s \cdot \frac{h_1(X)}{h_2(X)} + (1-s) \right) h_2(X) d\mu \\ &\geq \frac{3(1-s)}{2} \int_{\left\{ X \in \mathcal{X} : \left| \frac{h_1(X)}{h_2(X)} - \frac{1-s}{s} \right| \leq \frac{1-s}{s} \cdot \frac{2t}{1+t} \right\}} h_2(X) d\mu. \end{aligned}$$

The radius $\frac{2t}{1+t}$ here is not interchangeable with the $\frac{2t}{1-t}$ appearing in the upper bound, and the smaller of the two is the one that may be used. The set X_t is *asymmetric* about $\frac{1-s}{s}$: it reaches $\frac{1-s}{s} \cdot \frac{2t}{1-t}$ to the right but only $\frac{1-s}{s} \cdot \frac{2t}{1+t}$ to the left. A symmetric set of radius $\frac{1-s}{s} \cdot 2t$ is therefore *not* contained in X_t on the negative side, since $2t > \frac{2t}{1+t}$ for every $t > 0$, and so is not admissible in a lower bound; taking the radius to be the left endpoint itself makes the inclusion exact.

This costs nothing in the conclusion. Since $t \leq 1/4$ we have $\frac{2t}{1+t} \geq \frac{8t}{5}$, so the displayed integral dominates the same integral over the symmetric set of radius $\frac{1-s}{s} \cdot \frac{8t}{5}$, and a radius proportional to t is all that the $\mathcal{O}(t^q)$ conclusion requires. The proof of Lemma D.21 is complete. $\square$

With $C_q = \frac{c_0^{1/q}}{2q} (q+1)^{1+1/q}$ for $q > 0$ (at $q = 0$ the statement is trivial, since μ is a probability measure and any $C_0 \geq 1$ serves), Equation (3.4) follows after Lemma D.21 and Proposition 1 from [51].

D.2 Proof of Proposition 3.1

Proof of Proposition 3.1. Recall that in Proposition 3.1, we consider the domain $\mathcal{X}$ to be a union of two closed subdomains $\mathcal{X}_1$ and $\mathcal{X}_-$ such that the intersection of $\mathcal{X}_1$ and $\mathcal{X}_-$ is nonempty and has an empty interior. Suppose h_1 vanishes on $\mathcal{X}_-$ but its support equals $\mathcal{X}_1$ (i.e., $h_1 > 0$ on $\text{interior}(\mathcal{X}_1)$), and h_- vanishes on $\mathcal{X}_1$ but its support equals $\mathcal{X}_-$ (i.e., $h_- > 0$ on $\text{interior}(\mathcal{X}_-)$).

If no synthetic anomalies are generated, the resulting regression function f_P is given as

$$\begin{aligned} f_P(X) &= \frac{s \cdot h_1(X) - (1-s) \cdot h_-(X)}{s \cdot h_1(X) + (1-s) \cdot h_-(X)} \\ &= \begin{cases} \frac{h_1(X)}{h_1(X)} = 1, & \text{if } X \in \text{interior}(\mathcal{X}_1), \\ -\frac{h_-(X)}{h_-(X)} = -1, & \text{if } X \in \text{interior}(\mathcal{X}_-). \end{cases} \end{aligned}$$

In particular, we notice that f_P is discontinuous at the intersection of $\mathcal{X}_1$ and $\mathcal{X}_-$ (there is a jump from 1 to -1). Take an arbitrary point X^* in this intersection.

For any continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$, the function f takes its value $f(X^*)$ at X^* . In each case we approach X^* from the side on which f_P has the *opposite* sign to $f(X^*)$, which is what makes the gap at least 1. If $f(X^*) \geq 0$, we take a sequence $\{X^{(n)}\}_{n \in \mathbb{N}}$ in $\text{interior}(\mathcal{X}_-)$ tending to X^* , on which $f_P \equiv -1$, and then

$$\lim_{n \rightarrow \infty} \left(f(X^{(n)}) - f_P(X^{(n)}) \right) = f(X^*) - (-1) \geq 1.$$

In the same way, if $f(X^*) < 0$, we take a sequence $\{X^{(n)}\}_{n \in \mathbb{N}}$ in $\text{interior}(\mathcal{X}_1)$ tending to X^* , on which $f_P \equiv 1$, and then

$$\lim_{n \rightarrow \infty} \left(f(X^{(n)}) - f_P(X^{(n)}) \right) = f(X^*) - 1 < -1.$$

Both sequences exist because X^* lies in $\mathcal{X}_1 \cap \mathcal{X}_-$, and each $\mathcal{X}_i$ is the closure of its interior, being the support of a density that is positive there. Then $\|f - f_P\|_{L^\infty[0,1]^d} \geq 1$. This concludes the proof. $\square$

D.3 Proof of Proposition 4.5

Proof of Proposition 4.5. Write $A = sh_1 - (1-s)h_2$ and $B = sh_1 + (1-s)h_2$, so that $f_P = A/B$. Both h_1 and h_2 are non-negative, and $h_2 \geq 1 - \tilde{s} > 0$ by (4.1), so $B \geq (1-s)(1-\tilde{s}) > 0$ on $\mathcal{X}$ and $|A| \leq B$ pointwise.

Fix $x \neq y$ in $\mathcal{X}$. Writing the difference over a common denominator and adding and subtracting $A(y)B(y)$,

$$\begin{aligned} f_P(x) - f_P(y) &= \frac{A(x)B(y) - A(y)B(x)}{B(x)B(y)} \\ &= \frac{B(y)(A(x) - A(y)) + A(y)(B(y) - B(x))}{B(x)B(y)}, \end{aligned}$$

whence, using $|A(y)|/B(y) \leq 1$,

$$|f_P(x) - f_P(y)| \leq \frac{|A(x) - A(y)|}{B(x)} + \frac{|A(y)|}{B(y)} \cdot \frac{|B(x) - B(y)|}{B(x)} \leq \frac{[A]_\alpha + [B]_\alpha}{\inf_{\mathcal{X}} B} |x - y|^\alpha.$$

This is the first inequality of (4.2).

For the second, note that $h_2 = \tilde{s}h_- + (1 - \tilde{s}) \geq 1 - \tilde{s}$ since $h_- \geq 0$, so

$$\inf_{\mathcal{X}} B \geq (1 - s) \inf_{\mathcal{X}} h_2 \geq (1 - s)(1 - \tilde{s}).$$

The additive constant in h_2 has zero Hölder seminorm, so $[h_2]_\alpha \leq \tilde{s}[h_-]_\alpha$ and therefore

$$\max\{[A]_\alpha, [B]_\alpha\} \leq s[h_1]_\alpha + (1 - s)\tilde{s}[h_-]_\alpha.$$

Combining the three displays gives (4.2). $\square$

Remark D.22 (Sharpness, and the recovery of Proposition 3.1). *Both inequalities are tight in the regime of interest. Taking h_1 and h_- to be the disjointly supported hat functions of Example C.19 with $s = \frac{1}{2}$ and $\tilde{s} = 0$, a direct computation gives $[f_P]_1 = 31.9$ against a bound of 32.0, so the estimate is attained to within 0.4%.*

As $\tilde{s} \rightarrow 1$ the bound diverges, and it does so at the correct rate: for the same pair, $[f_P]_1$ grows proportionally to $(1 - \tilde{s})^{-1}$, and at $\tilde{s} = 1$ the regression function is the discontinuous step function of Proposition 3.1, whose Hölder seminorm is infinite. Proposition 4.5 is thus a quantitative refinement of Proposition 3.1 rather than a separate statement: the discontinuity there is the $\tilde{s} \rightarrow 1$ limit of the constant here.

E Proof of Theorem 3.3

We present the proof of Theorem 3.3 in this section.

Proof of Theorem 3.3. The k -order forward difference of a function g is defined as

$$\Delta_\tau^k g(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} g(x + j\tau),$$

which can be written as an integral sum

$$\Delta_\tau^k g(x) = \int_0^\tau \cdots \int_0^\tau g^{(k)}(x + t_1 + \cdots + t_k) dt_1 \cdots dt_k$$

Recall the ReLU^k function given by $\sigma^k(x) = (\max\{0, x\})^k$. Also recall the function σ_τ^k defined for some $0 < \tau \leq 1$ as

$$\sigma_\tau^k(x) = \frac{1}{k!\tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(x - j\tau) - \frac{1}{k!\tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(-x - j\tau).$$

Note that the k -th derivative of the ReLU^k function is

$$(\sigma^k)^{(k)}(x) = \begin{cases} k!, & \text{if } x > 0, \\ 0, & \text{if } x < 0. \end{cases}$$

If $x \geq 0$, we know $\sigma^k(-x - j\tau) = 0$ for every $j \in \mathbb{N}$ and $0 < \tau \leq 1$. Thus, for $x \geq 0$, we have

$$\begin{aligned}
\sigma_\tau^k(x) &= \frac{1}{k!\tau^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \sigma^k(x - j\tau) \\
&= \frac{1}{k!\tau^k} \sum_{j=0}^k (-1)^{k-j} \binom{k}{k-j} \sigma^k(x - k\tau + j\tau) \\
&= \frac{1}{k!\tau^k} \Delta_\tau^k(\sigma^k)(x - k\tau) \\
&= \frac{1}{k!\tau^k} \int_0^\tau \cdots \int_0^\tau (\sigma^k)^{(k)}(x - k\tau + t_1 + \cdots + t_k) dt_1 \cdots dt_k \\
&= \frac{1}{\tau^k} \int_0^\tau \cdots \int_0^\tau 1_{\{t_1 + \cdots + t_k > k\tau - x\}} dt_1 \cdots dt_k.
\end{aligned}$$

We can see from the above expression that $\sigma_\tau^k(0) = 0$, $\sigma_\tau^k(x) = 1$ for $x \geq k\tau$, and $\sigma_\tau^k(x)$ is strictly increasing for $x \in [0, k\tau]$. Similarly, for $x < 0$, we know that $\sigma_\tau^k(x) = -1$ for $x \leq -k\tau$, and $\sigma_\tau^k(x)$ is strictly increasing for $x \in [-k\tau, 0]$. Then,

$$\sigma_\tau^k(x) = \begin{cases} 1, & \text{if } x \geq k\tau, \\ -1, & \text{if } x \leq -k\tau \end{cases}$$

and $|\sigma_\tau^k(x)| \leq 1$ for $x \in (-k\tau, k\tau)$. Moreover, $\text{sign}(\sigma_\tau^k(x)) = \text{sign}(x)$.

Notice that the convex Hinge loss $\phi(x) = \max\{0, 1 - x\}$ is Lipschitz continuous on $\mathbb{R}$ with Lipschitz constant 1 because $|\phi(x_1) - \phi(x_2)| \leq |x_1 - x_2|$ for all $x_1, x_2 \in \mathbb{R}$.

For any function g such that $|g(X)| \leq 1$, we have $\phi(Yg(X)) = 1 - Yg(X)$, and the excess generalization error between such a g and the Bayes classifier f_c is

$$\begin{aligned}
\varepsilon(g) - \varepsilon(f_c) &= \int_{\mathcal{X}} \int_{\mathcal{Y}} 1 - Yg(X) - (1 - Yf_c(X)) dP(Y|X) dP_{\mathcal{X}} \\
&= \int_{\mathcal{X}} \int_{\mathcal{Y}} -Y(g(X) - f_c(X)) dP(Y|X) dP_{\mathcal{X}} \\
&= \int_{\mathcal{X}} (g(X) - f_c(X)) \int_{\mathcal{Y}} -Y dP(Y|X) dP_{\mathcal{X}} \\
&= \int_{\mathcal{X}} (f_c(X) - g(X)) (1 \cdot P(Y = 1|X) - 1 \cdot P(Y = -1|X)) dP_{\mathcal{X}} \\
&= \int_{\mathcal{X}} (f_c(X) - g(X)) (2\eta(X) - 1) dP_{\mathcal{X}}.
\end{aligned}$$

Notice that $2\eta(X) - 1 > 0$ if and only if $f_c(X) = \text{sign}(f_P(X)) = \text{sign}(2\eta(X) - 1)$, so we have

$$\begin{aligned}
\varepsilon(g) - \varepsilon(f_c) &= \int_{\mathcal{X}} |g(X) - f_c(X)| (2\eta(X) - 1) dP_{\mathcal{X}} \\
&= \int_{\mathcal{X}} |g(X) - f_c(X)| |f_P(X)| dP_{\mathcal{X}}.
\end{aligned}$$

Now take $g = \sigma_\tau^k(f)$. Let $\omega := \|f - f_P\|_{L^\infty[0,1]^d}$. We consider two cases separately: (i) $|f_P(X)| \leq \omega$ and (ii) $|f_P(X)| > \omega$. Note that

$$|\sigma_\tau^k(f(X)) - f_c(X)| \leq |\sigma_\tau^k(f(X))| + |f_c(X)| \leq 1 + 1 = 2, \quad \forall X \in \mathcal{X}. \quad (\text{E.1})$$

For the case $|f_P(X)| \leq \omega$, it follows from the Tsybakov noise condition (B.1) (with $q \in [0, \infty)$ and constant $c_0 > 0$) that

$$\begin{aligned}
& \int_{\{x \in \mathcal{X} : |f_P(X)| \leq \omega\}} |\sigma_\tau^k(f(X)) - f_c(X)| |f_P(X)| dP_{\mathcal{X}} \\
& \stackrel{(E.1)}{\leq} 2\omega \int_{\{x \in \mathcal{X} : |f_P(X)| \leq \omega\}} dP_{\mathcal{X}} \\
& = 2\omega \cdot \mathbb{P}_X(\{X \in \mathcal{X} : |f_P(X)| \leq \omega\}) \\
& \stackrel{(B.1)}{\leq} 2\omega \cdot c_0 \omega^q \\
& = 2c_0 \omega^{q+1}.
\end{aligned}$$

Next, for the other case $|f_P(X)| > \omega$, since $|f(X) - f_P(X)| \leq \omega$ for all $X \in \mathcal{X}$, we know

$$\text{sign}(\sigma_\tau^k(f(X))) = \text{sign}(f(X)) = \text{sign}(f_P(X)) = f_c(X).$$

If we have $\sigma_\tau^k(f(X)) \in \{-1, 1\}$, then

$$\left| \sigma_\tau^k(f(X)) - f_c(X) \right| = 0 \tag{E.2}$$

must hold. Also, if $|\sigma_\tau^k(f(X))| < 1$, then

$$|f(X)| < k\tau \tag{E.3}$$

and

$$|f_P(X)| = |f_P(X) - f(X) + f(X)| \leq \omega + |f(X)| \leq \omega + k\tau. \tag{E.4}$$

Thus, we have

$$\begin{aligned}
& \int_{\{x \in \mathcal{X} : |f_P(X)| > \omega\}} \left| \sigma_\tau^k(f(X)) - f_c(X) \right| |f_P(X)| dP_{\mathcal{X}} \\
& \stackrel{(E.2)}{=} \int_{\{x \in \mathcal{X} : |f_P(X)| > \omega, |\sigma_\tau^k(f(X))| < 1\}} \left| \sigma_\tau^k(f(X)) - f_c(X) \right| |f_P(X)| dP_{\mathcal{X}} \\
& \stackrel{(E.1), (E.4)}{\leq} 2(k\tau + \omega) \mathbb{P}_X \left(\left\{ x \in \mathcal{X} : |f_P(X)| > \omega, |\sigma_\tau^k(f(X))| < 1 \right\} \right) \\
& \leq 2(k\tau + \omega) \mathbb{P}_X \left(\left\{ x \in \mathcal{X} : |\sigma_\tau^k(f(X))| < 1 \right\} \right) \\
& \stackrel{(E.3)}{=} 2(k\tau + \omega) \mathbb{P}_X(\{x \in \mathcal{X} : |f(X)| < k\tau\}) \\
& \leq 2(k\tau + \omega) \mathbb{P}_X(\{x \in \mathcal{X} : |f_P(X)| < k\tau + \omega\}) \\
& \stackrel{(B.1)}{\leq} 2(k\tau + \omega) \cdot c_0 (k\tau + \omega)^q \\
& = 2c_0 (k\tau + \omega)^{q+1}.
\end{aligned}$$

Now combining the above error bounds, we get

$$\begin{aligned}
& \varepsilon \left(\sigma_\tau^k (f(X)) \right) - \varepsilon(f_c) \\
&= \int_{\mathcal{X}} \left| \sigma_\tau^k (f(X)) - f_c(X) \right| |f_P(X)| dP_{\mathcal{X}} \\
&= \int_{\{x \in \mathcal{X} : |f_P(X)| \leq \omega\}} \left| \sigma_\tau^k (f(X)) - f_c(X) \right| |f_P(X)| dP_{\mathcal{X}} \\
&\quad + \int_{\{x \in \mathcal{X} : |f_P(X)| > \omega\}} \left| \sigma_\tau^k (f(X)) - f_c(X) \right| |f_P(X)| dP_{\mathcal{X}} \\
&\leq 2c_0 \omega^{q+1} + 2c_0 (k\tau + \omega)^{q+1} \\
&\leq 4c_0 (k\tau + \omega)^{q+1} \\
&= 4c_0 \left(k\tau + \|f - f_P\|_{L^\infty[0,1]^d} \right)^{q+1}.
\end{aligned}$$

The proof is completed by the well-known comparison theorem in classification [58], which asserts that for the Hinge loss ϕ and any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$, the following inequality holds:

$$\underbrace{R(f) - R(f_c)}_{\text{excess risk}} \leq \underbrace{\varepsilon(f) - \varepsilon(f_c)}_{\text{excess generalization error}}. \quad (\text{E.5})$$

□

F Proof of Theorem 4.11

Let $T = \{(X_i, 1)\}_{i=1}^n$ be the set of real normal data given, where each X_i are drawn i.i.d. from an unknown distribution Q . Let $T^- = \{(X_i^-, -1)\}_{i=1}^{n^-}$ be the set of real anomalies given, where each X_i^- are drawn i.i.d. from another unknown distribution V with density h_- . Additionally, let $T' = \{(X'_i, -1)\}_{i=1}^{n'}$ be the set of synthetic anomalies generated from $\mu = \text{Uniform}(\mathcal{X})$. We merge these data together $T \cup T^- \cup T' = \{(X_i, 1)\}_{i=1}^n \cup \{(X_i^-, -1)\}_{i=1}^{n^-} \cup \{(X'_i, -1)\}_{i=1}^{n'}$ to train the ReLU network classifier. Theorem 4.11 proves that as the number of training data (n, n^-, n') increases, the ReLU network classifier achieves a theoretically grounded accuracy in anomaly detection.

F.1 Explicit excess risk bound

Before presenting the proof, we restate the theorem with an explicit excess risk bound, along with the corresponding parameters L^*, w^*, v^*, K^* .

Recall our hypothesis space $\mathcal{H}_\tau$ defined in Definition 4.9 with some $0 < \tau \leq 1$. Also recall the empirical risk minimizer w.r.t. Hinge loss ϕ trained on $T \cup T^- \cup T'$:

$$\begin{aligned}
& f_{\text{ERM}} \\
&= \arg \min_{f \in \mathcal{H}_\tau} \varepsilon_{T, T^-, T'}(f) \\
&= \arg \min_{f \in \mathcal{H}_\tau} \left[\frac{s}{n} \sum_{i=1}^n \phi(1 \cdot f(X_i)) + \frac{(1-s)\tilde{s}}{n^-} \sum_{i=1}^{n^-} \phi(-1 \cdot f(X_i^-)) + \frac{(1-s)(1-\tilde{s})}{n'} \sum_{i=1}^{n'} \phi(-1 \cdot f(X'_i)) \right].
\end{aligned}$$

Theorem F.23 (Restatement of Theorem 4.11). *Let $n, n^-, n' \geq 3, n_{\min} = \min\{n, n^-, n'\}, d \in \mathbb{N}, \alpha > 0$. Let $\ell \geq 1$ be an integer. Assume the Tsybakov noise condition holds for some noise exponent*

$q \in [0, \infty)$ and constant $c_0 > 0$. Consider the hypothesis space $\mathcal{H}_\tau$ with $N = \left\lceil \left(\frac{n_{\min}}{(\log(n_{\min}))^4} \right)^{\frac{d}{d+\alpha(q+2)}} \right\rceil$, $\tau = N^{-\frac{\alpha}{d}}$, $K^* = 1$, $L^* = 8 + (\ell + 5)(1 + \lceil \log_2(\max\{d, \alpha\}) \rceil)$, $w^* = 6(d + \lceil \alpha \rceil)N$, $v^* = 141(d + \alpha + 1)^{3+d}N(\ell + 6)$.

For any $0 < \delta < 1$, with probability $1 - \delta$, there holds,

$$R(\text{sign}(f_{\text{ERM}})) - R(f_c) \leq C \left(\log \left(\frac{6}{\delta} \max\{n, n^-, n'\} \right) \right) \left(\frac{(\log n_{\min})^4}{n_{\min}} \right)^{\frac{\alpha(q+1)}{d+\alpha(q+2)}},$$

where C is a positive constant independent of n or δ .

Remark F.24 (Where the main-text form comes from). The restatement above is the explicit bound the proof gives, and it carries a factor $\log\left(\frac{6}{\delta} \max\{n, n^-, n'\}\right)$ in the largest of the three sample sizes. The main-text statement of Theorem 4.11 is read in the smallest, and the polynomial comparability hypothesis stated there is exactly what licenses that. If $\max\{n, n^-, n'\} \leq (n_{\min})^c$ for a fixed $c \geq 1$, then, using $\log\left(\frac{6}{\delta}\right) \geq \log 6 > 1$ and $\log(n_{\min}) \geq \log 3 > 1$,

$$\log\left(\frac{6}{\delta} \max\{n, n^-, n'\}\right) \leq \log\left(\frac{6}{\delta}\right) + c \log(n_{\min}) \leq (1 + c) \log\left(\frac{6}{\delta}\right) \log(n_{\min}),$$

so the factor is absorbed into the $\tilde{O}$ and the rate depends on $n_{\min}$ alone. Without some such hypothesis it does not: holding $n_{\min}$ fixed and letting the largest sample size grow inflates the bound, which is why the informal statement's "only the smallest of the three sample sizes appears" is stated under the hypothesis rather than unconditionally. The same hypothesis, in the two-sample form $\max\{n, n^-\} \leq (\min\{n, n^-\})^c$, is what Zhou et al. [61] assume for their refined upper bound.

F.2 Error Decomposition

The well-known Comparison Theorem in classification [58] asserts that for the Hinge loss ϕ and any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$, the following inequality holds:

$$\underbrace{R(f) - R(f_c)}_{\text{excess risk}} \leq \underbrace{\varepsilon(f) - \varepsilon(f_c)}_{\text{excess generalization error}}. \quad (\text{F.1})$$

This result implies that, to establish an upper bound on the excess risk of a classifier f , it suffices to bound its excess generalization error, $\varepsilon(f) - \varepsilon(f_c)$.

The proof of Theorem 4.11 begins with a standard error decomposition of $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)$.

Lemma F.25 (Error Decomposition of $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)$). *Let $f_{\mathcal{H}}$ be any function in our hypothesis space $\mathcal{H}_\tau$. There holds*

$$\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \leq \{\varepsilon(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_{\text{ERM}})\} + \{\varepsilon_{T, T^-, T'}(f_{\mathcal{H}}) - \varepsilon(f_{\mathcal{H}})\} + \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}. \quad (\text{F.2})$$

The above lemma decomposes the excess generalization error $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)$ into three components. The first two components — $\{\varepsilon(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_{\text{ERM}})\}$ and $\{\varepsilon_{T, T^-, T'}(f_{\mathcal{H}}) - \varepsilon(f_{\mathcal{H}})\}$ — are commonly considered the estimation errors, whereas the last component $\{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}$ is the approximation error, which does not depend on the training data.

Moving forward, we will derive upper bounds for these three error terms, starting with the approximation error.

Proof of Lemma F.25. We express $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)$ by inserting empirical risks as follows

$$\begin{aligned} & \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \\ &= \{\varepsilon(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_{\text{ERM}})\} + \{\varepsilon_{T,T^-,T'}(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_{\mathcal{H}})\} \\ & \quad + \{\varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon(f_{\mathcal{H}})\} + \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}. \end{aligned}$$

We see that both f_{ERM} and $f_{\mathcal{H}}$ lie on $\mathcal{H}_{\tau}$. By the definition of f_{ERM} , we know that f_{ERM} minimizes the empirical risk over $\mathcal{H}_{\tau}$. Thus, we have $\varepsilon_{T,T^-,T'}(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) \leq 0$ for all $f_{\mathcal{H}} \in \mathcal{H}_{\tau}$. This yields the expression (F.2). $\square$

F.3 Bounding the Approximation Error

In this subsection, we focus on deriving an upper bound for the approximation error. To achieve this, we leverage a result from [42], which demonstrates that ReLU networks are capable of universally approximating any Hölder continuous function. For clarity, we first provide a formal definition of Hölder continuity.

We denote by $C^m(\mathcal{X})$ with $m \in \mathbb{N}$, the space of m -times differentiable functions on $\mathcal{X}$. Throughout this work, we consider the domain $\mathcal{X} = [0, 1]^d$; however, this can be extended to any compact subset of $\mathbb{R}^d$. For any positive value $\alpha > 0$, let $[\alpha]^- = \lceil \alpha \rceil - 1 \in \mathbb{N} \cup \{0\}$ be the largest integer strictly below α . Let $\beta = (\beta_1, \dots, \beta_d) \in \mathbb{N}_0^d$ be an index vector, where $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. We define $|\beta| = \beta_1 + \dots + \beta_d$ and $x^\beta = x_1^{\beta_1} \dots x_d^{\beta_d}$ for an index vector β . For a function $f : \mathcal{X} \rightarrow \mathbb{R}$ and an index vector $\beta \in \mathbb{N}_0^d$, let the partial derivative of f with β be

$$\partial^\beta f = \frac{\partial^{|\beta|} f}{\partial x^\beta} = \frac{\partial^{|\beta|} f}{\partial x_1^{\beta_1} \dots \partial x_d^{\beta_d}}.$$

The result from [42] specifically considers functions

$$f \in \mathcal{H}^{\alpha,r}([0, 1]^d) := \{f \in C^{[\alpha]^-}([0, 1]^d) : \|f\|_{\mathcal{H}^\alpha([0, 1]^d)} \leq r\}.$$

Here, $\mathcal{H}^{\alpha,r}([0, 1]^d)$ is a closed ball of radius $r > 0$ in the Hölder space of order $\alpha > 0$ w.r.t. the Hölder norm $\|\cdot\|_{\mathcal{H}^\alpha([0, 1]^d)}$ given by

$$\|f\|_{\mathcal{H}^\alpha([0, 1]^d)} = \sum_{|\beta|_1 \leq [\alpha]^-} \left\{ \|\partial^\beta f\|_{C([0, 1]^d)} + \sup_{x \neq y \in [0, 1]^d} \frac{|\partial^\beta f(x) - \partial^\beta f(y)|}{|x - y|^{\alpha - [\alpha]^-}} \right\}.$$

Lemma F.26 (Theorem 5 in [42]). *Let $\alpha, r > 0$. For any Hölder continuous function $f \in \mathcal{H}^{\alpha,r}([0, 1]^d)$ and for any integers $\ell \geq 1$ and $N \geq \max\{(\alpha + 1)^d, (r + 1)e^d\}$, there exists a ReLU neural network*

$$\widehat{f} \in \mathcal{F}(L^*, w^*, v^*, K^*)$$

with depth

$$L^* = 8 + (\ell + 5)(1 + \lceil \log_2(\max\{d, \alpha\}) \rceil),$$

maximum number of nodes

$$w^* = 6(d + \lceil \alpha \rceil)N,$$

number of nonzero parameters

$$v^* = 141(d + \alpha + 1)^{3+d}N(\ell + 6),$$

and all parameters (absolute value) are bounded by $K^ = 1$ such that*

$$\left\| \widehat{f} - f \right\|_{L^\infty([0, 1]^d)} \leq (2r + 1)(1 + d^2 + \alpha^2)6^d N 2^{-\ell} + r 3^\alpha N^{-\frac{\alpha}{d}}. \quad (\text{F.3})$$

We define our hypothesis space $\mathcal{H}_\tau$ based on the construction in Lemma F.26, with the goal of ensuring that its functions are well-suited to approximate the regression function f_P , assumed to be α -Hölder continuous. Recall the “approx-sign” function $\sigma_\tau : \mathbb{R} \rightarrow [-1, 1]$, defined previously in equation (3.7) with a bandwidth parameter $\tau > 0$. $\mathcal{H}_\tau$ is defined as

$$\mathcal{H}_\tau := \text{span} \{ \sigma_\tau \circ f : f \in \mathcal{F}(L^*, w^*, v^*, K^*) \}$$

with parameters L^*, w^*, v^*, K^* given in Lemma F.26, and τ to be determined later.

To estimate the term $\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)$, we apply the upper bound provided in Theorem 3.3 with $k = 1$. For any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$, there holds

$$\varepsilon(\sigma_\tau(f)) - \varepsilon(f_c) \leq 4c_0 \left(\tau + \|f - f_P\|_{L^\infty[0,1]^d} \right)^{q+1}.$$

Assuming $f_P \in \mathcal{H}^{\alpha,r}([0,1]^d)$ (i.e., f_P is α -Hölder continuous), we choose $f \in \mathcal{F}(L^*, w^*, v^*, K^*)$ such that

$$\|f - f_P\|_{L^\infty([0,1]^d)} \leq (2r+1)(1+d^2+\alpha^2)6^d N 2^{-\ell} + r 3^\alpha N^{-\frac{\alpha}{d}}.$$

We see that $\sigma_\tau(f) = f_{\mathcal{H}} \in \mathcal{H}_\tau$ and we have

$$\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c) \leq 4c_0 \left(\tau + (2r+1)(1+d^2+\alpha^2)6^d N 2^{-\ell} + r 3^\alpha N^{-\frac{\alpha}{d}} \right)^{q+1}.$$

We restrict $\tau \leq \min\{N 2^{-\ell} + N^{-\frac{\alpha}{d}}, 1\}$. We choose $c_{r,\alpha,d} = 4c_0(1+(2r+1)(1+d^2+\alpha^2)(6^d+3^\alpha))^{q+1}$. Then, we have

$$\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c) \leq c_{r,\alpha,d} \left(N 2^{-\ell} + N^{-\frac{\alpha}{d}} \right)^{q+1}. \quad (\text{F.4})$$

F.4 Bounding the Estimation Error

In this subsection, we focus on deriving an upper bound of the estimation error term $\{\varepsilon(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_{\text{ERM}})\} + \{\varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon(f_{\mathcal{H}})\}$.

We first rewrite this error term by inserting $\varepsilon(f_c)$ and $\varepsilon_{T,T^-,T'}(f_c)$ as

$$\begin{aligned} & \varepsilon(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_{\text{ERM}}) + \varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon(f_{\mathcal{H}}) \\ &= \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - \{\varepsilon_{T,T^-,T'}(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_c)\} \end{aligned} \quad (\text{F.5})$$

$$+ \varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon_{T,T^-,T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}. \quad (\text{F.6})$$

Next, we will bound the two terms (F.5) and (F.6) respectively. We will first handle the error term (F.6).

Step 1: Bound the term $\varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon_{T,T^-,T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}$.

Here, we define three random variables given by

$$A(X) := \phi(f_{\mathcal{H}}(X)) - \phi(f_c(X)) \quad \text{over } (\mathcal{X}, Q), \quad (\text{F.7})$$

and

$$B(X) := \phi(-f_{\mathcal{H}}(X)) - \phi(-f_c(X)) \quad \text{over } (\mathcal{X}, V), \quad (\text{F.8})$$

and

$$C(X) := \phi(-f_{\mathcal{H}}(X)) - \phi(-f_c(X)) \quad \text{over } (\mathcal{X}, \mu). \quad (\text{F.9})$$

Then, we can write $\varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon_{T,T^-,T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}$ as a function of these three random variables A, B , and C as follows:

$$\begin{aligned}
& \varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon_{T,T^-,T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\} \\
&= \frac{s}{n} \sum_{i=1}^n \phi(1 \cdot f_{\mathcal{H}}(X_i)) + \frac{(1-s)\tilde{s}}{n^-} \sum_{i=1}^{n^-} \phi(-1 \cdot f_{\mathcal{H}}(X_i^-)) + \frac{(1-s)(1-\tilde{s})}{n'} \sum_{i=1}^{n'} \phi(-1 \cdot f_{\mathcal{H}}(X'_i)) \\
&\quad - \left(\frac{s}{n} \sum_{i=1}^n \phi(1 \cdot f_c(X_i)) + \frac{(1-s)\tilde{s}}{n^-} \sum_{i=1}^{n^-} \phi(-1 \cdot f_c(X_i^-)) + \frac{(1-s)(1-\tilde{s})}{n'} \sum_{i=1}^{n'} \phi(-1 \cdot f_c(X'_i)) \right) \\
&\quad - \left(s \int_{\mathcal{X}} \phi(f_{\mathcal{H}}(X)) dQ + (1-s)\tilde{s} \int_{\mathcal{X}} \phi(-f_{\mathcal{H}}(X)) dW + (1-s)(1-\tilde{s}) \int_{\mathcal{X}} \phi(-f_{\mathcal{H}}(X)) d\mu \right) \\
&\quad + \left(s \int_{\mathcal{X}} \phi(f_c(X)) dQ + (1-s)\tilde{s} \int_{\mathcal{X}} \phi(-f_c(X)) dW + (1-s)(1-\tilde{s}) \int_{\mathcal{X}} \phi(-f_c(X)) d\mu \right) \\
&= s \left(\frac{1}{n} \sum_{i=1}^n A(X_i) - \mathbb{E}_Q[A(X)] \right) + (1-s)\tilde{s} \left(\frac{1}{n^-} \sum_{i=1}^{n^-} B(X_i^-) - \mathbb{E}_V[B(X)] \right) \\
&\quad + (1-s)(1-\tilde{s}) \left(\frac{1}{n'} \sum_{i=1}^{n'} C(X'_i) - \mathbb{E}_{\mu}[C(X)] \right).
\end{aligned}$$

Moving on, we will apply the classic one-sided Bernstein's inequality to estimate (i): $\frac{1}{n} \sum_{i=1}^n A(X_i) - \mathbb{E}_Q[A(X)]$, (ii): $\frac{1}{n^-} \sum_{i=1}^{n^-} B(X_i^-) - \mathbb{E}_V[B(X)]$, and (iii): $\frac{1}{n'} \sum_{i=1}^{n'} C(X'_i) - \mathbb{E}_{\mu}[C(X)]$, respectively, and obtain a high probability upper bound of $\varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon_{T,T^-,T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\}$. The result is given in the Lemma below:

Lemma F.27. *Suppose the Tsybakov noise condition holds for some $q \in [0, \infty)$ and constant $c_0 > 0$. For any $0 < \delta < 1$, with probability $1 - \frac{\delta}{2}$, we have*

$$\begin{aligned}
& \varepsilon_{T,T^-,T'}(f_{\mathcal{H}}) - \varepsilon_{T,T^-,T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\} \\
&\leq C_q \left(\log \left(\frac{6}{\delta} \right) \right) \left(s^{\frac{1}{q+2}} \left(\frac{1}{n} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot \tilde{s})^{\frac{1}{q+2}} \left(\frac{1}{n^-} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot (1-\tilde{s}))^{\frac{1}{q+2}} \left(\frac{1}{n'} \right)^{\frac{q+1}{q+2}} \right) \\
&\quad + \frac{(\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c))}{2},
\end{aligned} \tag{F.11}$$

where C_q is a positive constant depending only on q and c_0 .

Proof of Lemma F.27. The classic one-sided Bernstein's inequality states that, for a random variable A with mean $\mathbb{E}[A]$ and variance $\text{Var}[A] = \sigma^2$ satisfying $|A - \mathbb{E}[A]| \leq M$ almost surely (here $M = 2$, since ϕ takes values in $[0, 2]$ on $[-1, 1]$), the following holds for a random i.i.d. sample $\{X_i\}_{i=1}^n$ and any $t > 0$,

$$\mathbb{P} \left(\mathbb{E}[A] - \frac{1}{n} \sum_{i=1}^n A(X_i) > t \right) \leq \exp \left\{ -\frac{nt^2}{2(\sigma^2 + \frac{1}{3}Mt)} \right\}. \tag{F.12}$$

To apply Bernstein's inequality to random variables A (F.7), B (F.8), and C (F.9), we need to first derive upper bounds for their variances. A previous result in Lemma C.3 in [60] states that for any function $f : \mathcal{X} \rightarrow [-1, 1]$ and some $y \in \{-1, 1\}$, if the Tsybakov noise condition holds, we have

$$\mathbb{E}_{P_{\mathcal{X}}^y} [\{\phi(yf(X)) - \phi(yf_c(X))\}^2] \leq \frac{5}{s_y} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}, \tag{F.13}$$

where

$$s_y = s \quad \text{for } y = 1 \text{ and } s_y = 1 - s \quad \text{for } y = -1, \quad (\text{F.14})$$

and

$$dP_{\mathcal{X}}^y = h_1 d\mu \quad \text{for } y = 1 \text{ and } dP_{\mathcal{X}}^y = h_2 d\mu \quad \text{for } y = -1.$$

We can see that (F.13) presented an upper bound for the second moment of $\phi(yf(X)) - \phi(yf_c(X))$. Then, we know the variance of $\phi(yf(X)) - \phi(yf_c(X))$ is

$$\text{Var}[\phi(yf(X)) - \phi(yf_c(X))] \leq \frac{5}{s_y} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}.$$

For A this bounds the variance directly, since A is averaged over the sample T drawn from Q , whose density with respect to μ is exactly $h_1 = dP_{\mathcal{X}}^1/d\mu$:

$$\sigma_A^2 \leq \frac{5}{s} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}.$$

For B and C it does not, and this is the one place where our sampling scheme departs from the single-sample setting of [60]. The estimate (F.13) at $y = -1$ controls the second moment under $P_{\mathcal{X}}^{-1}$, whose density is h_2 — but B is averaged over T^- , drawn from h_- , and C over T' , drawn from μ , and neither sample is distributed as h_2 . The two are related by the mixture that defines the anomaly class,

$$h_2 = \tilde{s} h_- + (1 - \tilde{s}), \quad (\text{F.15})$$

so for any $g \geq 0$ both mixture components are dominated by the mixture,

$$\mathbb{E}_{h_2}[g] = \tilde{s} \mathbb{E}_{h_-}[g] + (1 - \tilde{s}) \mathbb{E}_{\mu}[g] \geq \max \{ \tilde{s} \mathbb{E}_{h_-}[g], (1 - \tilde{s}) \mathbb{E}_{\mu}[g] \},$$

and therefore

$$\begin{aligned} \sigma_B^2 &\leq \frac{1}{\tilde{s}} \cdot \frac{5}{1-s} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} = \frac{5}{(1-s)\tilde{s}} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}, \\ \sigma_C^2 &\leq \frac{1}{1-\tilde{s}} \cdot \frac{5}{1-s} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} = \frac{5}{(1-s)(1-\tilde{s})} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}. \end{aligned}$$

The factors $\tilde{s}^{-1}$ and $(1 - \tilde{s})^{-1}$ are not cosmetic: each is exactly cancelled by the weight $(1 - s)\tilde{s}$, respectively $(1 - s)(1 - \tilde{s})$, that the corresponding term carries in $\varepsilon_{T, T^-, T'}$, and it is that cancellation which produces the symmetric factors $((1 - s)\tilde{s})^{\frac{1}{q+2}}$ and $((1 - s)(1 - \tilde{s}))^{\frac{1}{q+2}}$ in the statement of the lemma.

Now we are in a position to apply the one-sided Bernstein's inequality (F.12). For any $t > 0$, there holds, with probability at least $1 - \exp\left(-\frac{nt^2}{2(\sigma^2 + 2t/3)}\right)$,

$$\left(\frac{1}{n} \sum_{i=1}^n A(X_i) - \mathbb{E}_Q[A(X)] \right) \leq t.$$

For $0 < \delta < 1$, we set

$$1 - \exp\left(-\frac{nt^2}{2(\sigma^2 + 2t/3)}\right) = 1 - \frac{\delta}{6},$$

and we solve t for the quadratic equation:

$$\log\left(\frac{6}{\delta}\right) = \frac{nt^2}{2(\sigma^2 + 2t/3)}.$$

The positive solution to this quadratic equation of t is given by

$$\begin{aligned}
t^* &= \frac{\frac{4}{3} \log\left(\frac{6}{\delta}\right) + \sqrt{\frac{16}{9} \log^2\left(\frac{6}{\delta}\right) + 8n\sigma^2 \log\left(\frac{6}{\delta}\right)}}{2n} \\
&\leq \frac{2}{3n} \log\left(\frac{6}{\delta}\right) + \frac{2}{3n} \log\left(\frac{6}{\delta}\right) + \frac{\sqrt{2\sigma^2 \log\left(\frac{6}{\delta}\right)}}{\sqrt{n}} \\
&\leq \frac{4}{3n} \log\left(\frac{6}{\delta}\right) + 4 \frac{\sqrt{\log\left(\frac{6}{\delta}\right)}}{\sqrt{ns}} (c_0)^{\frac{1}{2(q+1)}} (\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c))^{\frac{q}{2(q+1)}}.
\end{aligned}$$

We further apply Young's inequality for products [57] to the above estimate of t^* , and we get

$$t^* \leq \frac{4}{3n} \log\left(\frac{6}{\delta}\right) + \left(\frac{q+2}{2(q+1)}\right) \left(4 \frac{\sqrt{\log\left(\frac{6}{\delta}\right)}}{\sqrt{ns}} (c_0)^{\frac{1}{2(q+1)}}\right)^{\frac{2(q+1)}{q+2}} + \frac{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)}{\frac{2(q+1)}{q}}.$$

In other words, we know for probability at least $1 - \frac{\delta}{6}$, there holds

$$\begin{aligned}
&\left(\frac{1}{n} \sum_{i=1}^n A(X_i) - \mathbb{E}_Q[A(X)]\right) \\
&\leq \frac{4}{3n} \log\left(\frac{6}{\delta}\right) + \left(\frac{q+2}{2(q+1)}\right) \left(4 \frac{\sqrt{\log\left(\frac{6}{\delta}\right)}}{\sqrt{ns}} (c_0)^{\frac{1}{2(q+1)}}\right)^{\frac{2(q+1)}{q+2}} + \frac{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)}{\frac{2(q+1)}{q}}.
\end{aligned}$$

We adopt the same approach and we obtain that, with probability at least $1 - \frac{\delta}{6}$, there holds

$$\begin{aligned}
&\left(\frac{1}{n^-} \sum_{i=1}^{n^-} B(X_i^-) - \mathbb{E}_V[B(X^-)]\right) \\
&\leq \frac{4}{3n^-} \log\left(\frac{6}{\delta}\right) + \left(\frac{q+2}{2(q+1)}\right) \left(4 \frac{\sqrt{\log\left(\frac{6}{\delta}\right)}}{\sqrt{n^-(1-s)\tilde{s}}} (c_0)^{\frac{1}{2(q+1)}}\right)^{\frac{2(q+1)}{q+2}} + \frac{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)}{\frac{2(q+1)}{q}}
\end{aligned}$$

and

$$\begin{aligned}
&\left(\frac{1}{n'} \sum_{i=1}^{n'} C(X'_i) - \mathbb{E}_{\mu}[C(X')]\right) \\
&\leq \frac{4}{3n'} \log\left(\frac{6}{\delta}\right) + \left(\frac{q+2}{2(q+1)}\right) \left(4 \frac{\sqrt{\log\left(\frac{6}{\delta}\right)}}{\sqrt{n'(1-s)(1-\tilde{s})}} (c_0)^{\frac{1}{2(q+1)}}\right)^{\frac{2(q+1)}{q+2}} + \frac{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)}{\frac{2(q+1)}{q}}.
\end{aligned}$$

Combining the above estimates, we know, for probability at least $1 - \frac{\delta}{2}$,

$$\begin{aligned}
& \varepsilon_{T, T^-, T'}(f_{\mathcal{H}}) - \varepsilon_{T, T^-, T'}(f_c) - \{\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)\} \\
&= s \left(\frac{1}{n} \sum_{i=1}^n A(X_i) - \mathbb{E}_Q[A(X)] \right) + (1-s)\tilde{s} \left(\frac{1}{n^-} \sum_{i=1}^{n^-} B(X_i^-) - \mathbb{E}_V[B(X)] \right) \\
&\quad + (1-s)(1-\tilde{s}) \left(\frac{1}{n'} \sum_{i=1}^{n'} C(X'_i) - \mathbb{E}_\mu[C(X)] \right) \\
&\leq \frac{4}{3} \left(\frac{s}{n} + \frac{(1-s)\tilde{s}}{n^-} + \frac{(1-s)(1-\tilde{s})}{n'} \right) \log \left(\frac{6}{\delta} \right) \\
&\quad + 4^{\frac{2(q+1)}{q+2}} (c_0)^{\frac{1}{q+2}} \left(\log \left(\frac{6}{\delta} \right) \right)^{\frac{q+1}{q+2}} \\
&\quad \cdot \left(s^{\frac{1}{q+2}} \left(\frac{1}{n} \right)^{\frac{q+1}{q+2}} + ((1-s)\tilde{s})^{\frac{1}{q+2}} \left(\frac{1}{n^-} \right)^{\frac{q+1}{q+2}} + ((1-s)(1-\tilde{s}))^{\frac{1}{q+2}} \left(\frac{1}{n'} \right)^{\frac{q+1}{q+2}} \right) \\
&\quad + \left(\frac{q}{2(q+1)} \right) (\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)) \\
&\leq C_q \left(\log \left(\frac{6}{\delta} \right) \right) \\
&\quad \cdot \left(s^{\frac{1}{q+2}} \left(\frac{1}{n} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot \tilde{s})^{\frac{1}{q+2}} \left(\frac{1}{n^-} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot (1-\tilde{s}))^{\frac{1}{q+2}} \left(\frac{1}{n'} \right)^{\frac{q+1}{q+2}} \right) \\
&\quad + \frac{(\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c))}{2},
\end{aligned}$$

where C_q is a positive constant depending only on q and c_0 . The mixture weights need no further relaxation: each of the three second terms already carries its own weight to the power $\frac{1}{q+2}$, because the weight $(1-s)\tilde{s}$ multiplying the B term cancels against the $((1-s)\tilde{s})^{-\frac{q+1}{q+2}}$ produced by its variance bound, leaving $((1-s)\tilde{s})^{1-\frac{q+1}{q+2}} = ((1-s)\tilde{s})^{\frac{1}{q+2}}$, and likewise for A and C . In the last inequality we have dropped the factor $\frac{q+2}{2(q+1)} \leq 1$, valid for every $q \geq 0$, and used the fact that $\frac{1}{n} \leq \left(\frac{1}{n}\right)^{\frac{q+1}{q+2}}$, $\frac{1}{n^-} \leq \left(\frac{1}{n^-}\right)^{\frac{q+1}{q+2}}$, $\frac{1}{n'} \leq \left(\frac{1}{n'}\right)^{\frac{q+1}{q+2}}$ for all $n, n^-, n' \in \mathbb{N}$ and $q > 0$. $\square$

Step 2: Bound the term $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - \{\varepsilon_{T, T^-, T'}(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_c)\}$.

Here, we will proceed to estimate the error term $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - \{\varepsilon_{T, T^-, T'}(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_c)\}$, which is given in (F.5). We will derive an upper bound for this error term using a concentration inequality in terms of covering numbers.

For $\epsilon > 0$, denote by $\mathcal{N}(\epsilon, \mathcal{H}) := \mathcal{N}(\epsilon, \mathcal{H}, \|\cdot\|_\infty)$ the ϵ -covering number of a set of functions $\mathcal{H}$ with respect to $\|\cdot\|_\infty := \text{ess sup}_{x \in \mathcal{X}} |f(x)|$. More specifically, $\mathcal{N}(\epsilon, \mathcal{H})$ is the minimal $M \in \mathbb{N}$ such that there exist functions $\{f_1, \dots, f_M\} \in \mathcal{H}$ satisfying

$$\min_{1 \leq i \leq M} \|f - f_i\|_\infty \leq \epsilon, \quad \forall f \in \mathcal{H}. \quad (\text{F.16})$$

In particular, we seek to estimate the covering number of our hypothesis space $\mathcal{H}_\tau$ defined in Definition 4.9. $\mathcal{H}_\tau$ is the class of ReLU neural networks we use to learn the Bayes classifier.

Lemma F.28 (Corollary C.6 from [60]). *Consider the hypothesis space $\mathcal{H}_\tau$ defined in Definition 4.9 with hyperparameter $0 < \tau \leq 1$, and integers $\ell \geq 1$ and $N \geq \max\{(\alpha + 1)^d, (r + 1)e^d\}$. For any $0 < \epsilon \leq 1$, the ϵ -covering number of $\mathcal{H}_\tau$ satisfies*

$$\log \mathcal{N}(\epsilon, \mathcal{H}_\tau) \leq c_{\alpha,d} \ell^2 N \log((\tau\epsilon)^{-1} \ell N), \quad (\text{F.17})$$

where $c_{\alpha,d}$ is a positive constant independent of r, ℓ, N, τ or ϵ .

Next, by utilizing this estimate of the ϵ -covering number of $\mathcal{H}_\tau$, we derive the following upper bound for the error term $\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - \{\varepsilon_{T,T^-,T'}(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_c)\}$.

Lemma F.29. *Let $\alpha, r > 0$, and integers $\ell \geq 1$ and $N \geq \max\{(\alpha + 1)^d, (r + 1)e^d\}$. Suppose the Tsybakov noise condition (3.3) holds for some $0 \leq q < \infty$ and constant $c_0 > 0$. Also let $\tau \geq \max\{\frac{1}{n}, \frac{1}{n^-}, \frac{1}{n'}\}$. For any $0 < \delta < 1$ and $n, n^-, n' \geq 3$, with probability $1 - \frac{\delta}{2}$, we have*

$$\begin{aligned} & \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - (\varepsilon_{T,T^-,T'}(f_{\text{ERM}}) - \varepsilon_{T,T^-,T'}(f_c)) \\ & \leq C_{q,\alpha,d} \left(\log\left(\frac{6}{\delta}\right) + 3\ell^2 N \log(\max\{n, n^-, n'\} \ell N) \right)^{\frac{q+1}{q+2}} \\ & \quad \cdot \left(\max\left\{ \frac{s}{n} \log\left(\frac{n}{s}\right), \frac{(1-s)\tilde{s}}{n^-} \log\left(\frac{n^-}{(1-s)\tilde{s}}\right), \frac{(1-s)(1-\tilde{s})}{n'} \log\left(\frac{n'}{(1-s)(1-\tilde{s})}\right) \right\} \right)^{\frac{q+1}{q+2}} \\ & \quad + \frac{\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)}{2}, \end{aligned}$$

where $C_{q,\alpha,d}$ is a positive constant depending only on q, c_0, α , and d .

Proof of Lemma F.29. Consider a fixed function $f : \mathcal{X} \rightarrow [-1, 1]$. For $s, \tilde{s} \in (0, 1)$, we apply the one-sided Bernstein's inequality (F.12) to following three random variables:

$$a(X) = s(\phi(f_{\mathcal{H}}(X)) - \phi(f_c(X))) \quad \text{over } (\mathcal{X}, Q),$$

and

$$b(X) = (1-s)\tilde{s}(\phi(-f_{\mathcal{H}}(X)) - \phi(-f_c(X))) \quad \text{over } (\mathcal{X}, V),$$

and

$$c(X) = (1-s)(1-\tilde{s})(\phi(-f_{\mathcal{H}}(X)) - \phi(-f_c(X))) \quad \text{over } (\mathcal{X}, \mu).$$

We can see that, almost surely, $|a| \leq 2s$ and thereby $|a - \mathbb{E}[a]| \leq 4s$, $|b| \leq 2(1-s)\tilde{s}$ and thereby $|b - \mathbb{E}[b]| \leq 4(1-s)\tilde{s}$, and $|c| \leq 2(1-s)(1-\tilde{s})$ and thereby $|c - \mathbb{E}[c]| \leq 4(1-s)(1-\tilde{s})$. Moreover, by applying the argument in the proof of Lemma F.27 — *including* the mixture domination (F.15), which contributes $\tilde{s}^{-1}$ to $\text{Var}[b]$ and $(1-\tilde{s})^{-1}$ to $\text{Var}[c]$ because b is averaged over h_- and c over μ , neither of which is h_2 — we know that when the noise condition holds for some $0 \leq q < \infty$, the variances of these random variables can be bounded by

$$\text{Var}[a] \leq \frac{5s^2}{s} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} = 5s(c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}$$

and

$$\text{Var}[b] \leq \frac{5((1-s)\tilde{s})^2}{(1-s)\tilde{s}} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} = 5(1-s)\tilde{s} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}$$

and

$$\text{Var}[c] \leq \frac{5((1-s)(1-\tilde{s}))^2}{(1-s)(1-\tilde{s})} (c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} = 5(1-s)(1-\tilde{s})(c_0)^{\frac{1}{q+1}} (\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}}.$$

In each case the squared prefactor of the random variable is divided by the mixture weight of the sample it is averaged over, and the three bounds are exactly what the denominators of the Bernstein exponents below require. We then apply the one-sided Bernstein's inequality to each of these random variables. We combine the estimates together and get

$$\frac{\varepsilon(f) - \varepsilon(f_c) - (\varepsilon_{T,T^-,T'}(f) - \varepsilon_{T,T^-,T'}(f_c))}{\left((\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} + \epsilon^{\frac{q}{q+1}}\right)^{1/2}} \leq 2\epsilon^{1-\frac{q}{2(q+1)}}, \quad \forall \epsilon > 0$$

with probability at least

$$\begin{aligned} 1 - \exp \left\{ -\frac{n\epsilon^{2-\frac{q}{q+1}}}{s \left(10(c_0)^{\frac{1}{q+1}} + 2\epsilon^{1-\frac{q}{q+1}}\right)} \right\} &= \exp \left\{ -\frac{n^- \epsilon^{2-\frac{q}{q+1}}}{(1-s)\tilde{s} \left(10(c_0)^{\frac{1}{q+1}} + 2\epsilon^{1-\frac{q}{q+1}}\right)} \right\} \\ &= \exp \left\{ -\frac{n' \epsilon^{2-\frac{q}{q+1}}}{(1-s)(1-\tilde{s}) \left(10(c_0)^{\frac{1}{q+1}} + 2\epsilon^{1-\frac{q}{q+1}}\right)} \right\}. \end{aligned}$$

Specifically, we consider our hypothesis space $\mathcal{H}_\tau$ and its covering number estimate $\mathcal{N}(\epsilon, \mathcal{H}_\tau)$, as given in (F.17) for any $\epsilon > 0$. Building on a similar result from Lemma C.8 in [60], if the noise condition holds for some $0 \leq q < \infty$, we have for any $\epsilon > 0$,

$$\varepsilon(f) - \varepsilon(f_c) - (\varepsilon_{T,T^-,T'}(f) - \varepsilon_{T,T^-,T'}(f_c)) \leq 5\epsilon^{1-\frac{q}{2(q+1)}} \left((\varepsilon(f) - \varepsilon(f_c))^{\frac{q}{q+1}} + \epsilon^{\frac{q}{q+1}} \right)^{\frac{1}{2}}, \quad \forall f \in \mathcal{H}_\tau$$

with probability at least

$$\begin{aligned} &1 - \mathcal{N}(\epsilon, \mathcal{H}_\tau) \left\{ \exp \left\{ -\frac{n\epsilon^{2-\frac{q}{q+1}}}{s \left(10(c_0)^{\frac{1}{q+1}} + 2\epsilon^{1-\frac{q}{q+1}}\right)} \right\} \right. \\ &\quad + \exp \left\{ -\frac{n^- \epsilon^{2-\frac{q}{q+1}}}{(1-s)\tilde{s} \left(10(c_0)^{\frac{1}{q+1}} + 2\epsilon^{1-\frac{q}{q+1}}\right)} \right\} \\ &\quad \left. + \exp \left\{ -\frac{n' \epsilon^{2-\frac{q}{q+1}}}{(1-s)(1-\tilde{s}) \left(10(c_0)^{\frac{1}{q+1}} + 2\epsilon^{1-\frac{q}{q+1}}\right)} \right\} \right\} \\ &\geq 1 - \exp\{c_{\alpha,d}\ell^2 N \log((\tau\epsilon)^{-1}\ell N)\} \\ &\quad \left\{ \exp \left\{ -\frac{n\epsilon^{2-\frac{q}{q+1}}}{s \left(10(c_0)^{\frac{1}{q+1}} + 2\right)} \right\} + \exp \left\{ -\frac{n^- \epsilon^{2-\frac{q}{q+1}}}{(1-s)\tilde{s} \left(10(c_0)^{\frac{1}{q+1}} + 2\right)} \right\} \right. \\ &\quad \left. + \exp \left\{ -\frac{n' \epsilon^{2-\frac{q}{q+1}}}{(1-s)(1-\tilde{s}) \left(10(c_0)^{\frac{1}{q+1}} + 2\right)} \right\} \right\}. \end{aligned}$$

We choose τ satisfying $\tau \geq \max\{\frac{1}{n}, \frac{1}{n^-}, \frac{1}{n'}\}$. Then we know $\tau^{-1} \leq n$, $\tau^{-1} \leq n^-$, and $\tau^{-1} \leq n'$. We set the above confidence bound to be at least $1 - \delta/2$. Then, we find $\epsilon > 0$ that satisfies the

following three inequalities:

$$\exp \left\{ c_{\alpha,d} \ell^2 N \log(\epsilon^{-1} n \ell N) - \frac{n \epsilon^{2-\frac{q}{q+1}}}{s \left(10(c_0)^{\frac{1}{q+1}} + 2 \right)} \right\} \leq \frac{\delta}{6}$$

and

$$\exp \left\{ c_{\alpha,d} \ell^2 N \log(\epsilon^{-1} n^- \ell N) - \frac{n^- \epsilon^{2-\frac{q}{q+1}}}{(1-s)\tilde{s} \left(10(c_0)^{\frac{1}{q+1}} + 2 \right)} \right\} \leq \frac{\delta}{6}$$

and

$$\exp \left\{ c_{\alpha,d} \ell^2 N \log(\epsilon^{-1} n' \ell N) - \frac{n' \epsilon^{2-\frac{q}{q+1}}}{(1-s)(1-\tilde{s}) \left(10(c_0)^{\frac{1}{q+1}} + 2 \right)} \right\} \leq \frac{\delta}{6}.$$

Following the proof of Lemma C.8 in [60], we know ϵ satisfying all the above three inequalities can be given by

$$\epsilon = \left(\max \left\{ \frac{Bs \log \left(\frac{n}{s} \right)}{n}, \frac{B^-(1-s)\tilde{s}}{n^-} \log \left(\frac{n^-}{(1-s)\tilde{s}} \right), \frac{B'(1-s)(1-\tilde{s})}{n'} \log \left(\frac{n'}{(1-s)(1-\tilde{s})} \right) \right\} \right)^{\frac{q+1}{q+2}}, \quad (\text{F.18})$$

where $c_q = 10(c_0)^{\frac{1}{q+1}} + 2$, $B := c_q \left(\frac{2c_{\alpha,d}}{2-\frac{q}{q+1}} \ell^2 N + \log \left(\frac{6}{\delta} \right) + c_{\alpha,d} \ell^2 N \log(n \ell N) \right)$,

$B^- := c_q \left(\frac{2c_{\alpha,d}}{2-\frac{q}{q+1}} \ell^2 N + \log \left(\frac{6}{\delta} \right) + c_{\alpha,d} \ell^2 N \log(n^- \ell N) \right)$, and

$B' := c_q \left(\frac{2c_{\alpha,d}}{2-\frac{q}{q+1}} \ell^2 N + \log \left(\frac{6}{\delta} \right) + c_{\alpha,d} \ell^2 N \log(n' \ell N) \right)$. Note that

$$B \leq c_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + \ell^2 N (1 + \log(n \ell N)) \right), \quad (\text{F.19})$$

where $c_{q,\alpha,d} = c_q \left(\frac{2c_{\alpha,d}}{2-\frac{q}{q+1}} + c_{\alpha,d} \right) \geq 1$ is a positive constant depending only on q, c_0, α , and d . We also note that

$$B^- \leq c_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + \ell^2 N (1 + \log(n^- \ell N)) \right) \quad (\text{F.20})$$

and

$$B' \leq c_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + \ell^2 N (1 + \log(n' \ell N)) \right). \quad (\text{F.21})$$

Next, we take $f = f_{\text{ERM}} \in \mathcal{H}_\tau$. We take ϵ chosen in (F.18). We apply Young's inequality for

products [57], and with probability at least $1 - \frac{\delta}{2}$,

$$\begin{aligned}
& \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - (\varepsilon_{T, T^-, T'}(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_c)) \\
& \leq 5\epsilon^{1 - \frac{q}{2(q+1)}} \left((\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c))^{\frac{q}{q+1}} + \epsilon^{\frac{q}{q+1}} \right)^{\frac{1}{2}} \\
& \leq 5^{\frac{2(q+1)}{q+2}} \\
& \quad \cdot \left(\max \left\{ \frac{Bs \log\left(\frac{n}{s}\right)}{n}, \frac{B^-(1-s)\tilde{s}}{n^-} \log\left(\frac{n^-}{(1-s)\tilde{s}}\right), \frac{B'(1-s)(1-\tilde{s})}{n'} \log\left(\frac{n'}{(1-s)(1-\tilde{s})}\right) \right\} \right)^{\frac{q+1}{q+2}} \\
& \quad + \frac{\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)}{2} \\
& \leq 5^{\frac{2(q+1)}{q+2}} (\max\{B, B^-, B'\})^{\frac{q+1}{q+2}} \\
& \quad \cdot \left(\max \left\{ \frac{s \log\left(\frac{n}{s}\right)}{n}, \frac{(1-s)\tilde{s}}{n^-} \log\left(\frac{n^-}{(1-s)\tilde{s}}\right), \frac{(1-s)(1-\tilde{s})}{n'} \log\left(\frac{n'}{(1-s)(1-\tilde{s})}\right) \right\} \right)^{\frac{q+1}{q+2}} \\
& \quad + \frac{\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)}{2}.
\end{aligned}$$

We then apply the upper bounds of B (F.19), B^- (F.20), and B' (F.21) and get, with probability at least $1 - \frac{\delta}{2}$,

$$\begin{aligned}
& \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) - (\varepsilon_{T, T^-, T'}(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_c)) \\
& \leq C_{q, \alpha, d} \left(\log\left(\frac{6}{\delta}\right) + \ell^2 N (\log(n\ell N) + \log(n^-\ell N) + \log(n'\ell N)) \right)^{\frac{q+1}{q+2}} \\
& \quad \cdot \left(\max \left\{ \frac{s \log\left(\frac{n}{s}\right)}{n}, \frac{(1-s)\tilde{s}}{n^-} \log\left(\frac{n^-}{(1-s)\tilde{s}}\right), \frac{(1-s)(1-\tilde{s})}{n'} \log\left(\frac{n'}{(1-s)(1-\tilde{s})}\right) \right\} \right)^{\frac{q+1}{q+2}} \\
& \quad + \frac{\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)}{2} \\
& \leq C_{q, \alpha, d} \left(\log\left(\frac{6}{\delta}\right) + 3\ell^2 N \log(\max\{n, n^-, n'\}\ell N) \right)^{\frac{q+1}{q+2}} \\
& \quad \cdot \left(\max \left\{ \frac{s \log\left(\frac{n}{s}\right)}{n}, \frac{(1-s)\tilde{s}}{n^-} \log\left(\frac{n^-}{(1-s)\tilde{s}}\right), \frac{(1-s)(1-\tilde{s})}{n'} \log\left(\frac{n'}{(1-s)(1-\tilde{s})}\right) \right\} \right)^{\frac{q+1}{q+2}} \\
& \quad + \frac{\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)}{2},
\end{aligned}$$

where $C_{q, \alpha, d}$ is a positive constant depending only on q, c_0, α , and d . □

F.5 Combining error bounds together

Assume $n, n^-, n' \geq 3$. Let $\tau \geq \max\{\frac{1}{n}, \frac{1}{n^-}, \frac{1}{n'}\}$ and $\tau \leq \min\{N2^{-\ell} + N^{-\frac{\alpha}{d}}, 1\}$. Suppose the Tsybakov noise condition holds for some $q \in [0, \infty)$ and constant $c_0 > 0$. For any $0 < \delta < 1$, with

probability at least $1 - \delta$, we have

$$\begin{aligned}
& \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \\
& \leq \{ \varepsilon(f_{\text{ERM}}) - \varepsilon_{T, T^-, T'}(f_{\text{ERM}}) \} + \{ \varepsilon_{T, T^-, T'}(f_{\mathcal{H}}) - \varepsilon(f_{\mathcal{H}}) \} + \{ \varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c) \} \\
& \leq C_{q, \alpha, d} \left(\log \left(\frac{6}{\delta} \right) + 3\ell^2 N \log(\max\{n, n^-, n'\} \ell N) \right)^{\frac{q+1}{q+2}} \\
& \quad \cdot \left(\max \left\{ \frac{s \log \left(\frac{n}{s} \right)}{n}, \frac{(1-s)\tilde{s}}{n^-} \log \left(\frac{n^-}{(1-s)\tilde{s}} \right), \frac{(1-s)(1-\tilde{s})}{n'} \log \left(\frac{n'}{(1-s)(1-\tilde{s})} \right) \right\} \right)^{\frac{q+1}{q+2}} \\
& \quad + \frac{\varepsilon(f_{\text{ERM}}) - \varepsilon(f_c)}{2} \\
& \quad + C_q \left(\log \left(\frac{6}{\delta} \right) \right) \\
& \quad \cdot \left(s^{\frac{1}{q+2}} \left(\frac{1}{n} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot \tilde{s})^{\frac{1}{q+2}} \left(\frac{1}{n^-} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot (1-\tilde{s}))^{\frac{1}{q+2}} \left(\frac{1}{n'} \right)^{\frac{q+1}{q+2}} \right) \\
& \quad + \frac{(\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c))}{2} + (\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)),
\end{aligned}$$

where we recall that $C_{q, \alpha, d}$ is a positive constant depending only on q, c_0, α , and d ; and C_q is a positive constant depending only on q and c_0 .

We multiply both sides of the inequality by 2 and substitute the estimate of the final term, $\varepsilon(f_{\mathcal{H}}) - \varepsilon(f_c)$ (i.e., the approximation error term), from (F.4). This yields

$$\begin{aligned}
& \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \\
& \leq 2C_{q, \alpha, d} \left(\log \left(\frac{6}{\delta} \right) + 3\ell^2 N \log(\max\{n, n^-, n'\} \ell N) \right)^{\frac{q+1}{q+2}} \\
& \quad \cdot \left(\max \left\{ \frac{s \log \left(\frac{n}{s} \right)}{n}, \frac{(1-s)\tilde{s}}{n^-} \log \left(\frac{n^-}{(1-s)\tilde{s}} \right), \frac{(1-s)(1-\tilde{s})}{n'} \log \left(\frac{n'}{(1-s)(1-\tilde{s})} \right) \right\} \right)^{\frac{q+1}{q+2}} \\
& \quad + 2C_q \left(\log \left(\frac{6}{\delta} \right) \right) \\
& \quad \cdot \left(s^{\frac{1}{q+2}} \left(\frac{1}{n} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot \tilde{s})^{\frac{1}{q+2}} \left(\frac{1}{n^-} \right)^{\frac{q+1}{q+2}} + ((1-s) \cdot (1-\tilde{s}))^{\frac{1}{q+2}} \left(\frac{1}{n'} \right)^{\frac{q+1}{q+2}} \right) \\
& \quad + 3c_{r, \alpha, d} \left(N2^{-\ell} + N^{-\frac{\alpha}{d}} \right)^{q+1}.
\end{aligned}$$

We select the smallest $\ell \in \mathbb{N}$ such that $N2^{-\ell} \leq N^{-\frac{\alpha}{d}}$. This condition translates to:

$$\ell = \left\lceil \left(1 + \frac{\alpha}{d} \right) \frac{\log N}{\log 2} \right\rceil.$$

With this choice of ℓ , it follows that:

$$N2^{-\ell} + N^{-\frac{\alpha}{d}} \leq 2N^{-\frac{\alpha}{d}}.$$

We also know that

$$\frac{s}{n} \log \left(\frac{n}{s} \right) \leq \frac{\log n}{n}$$

and

$$\frac{(1-s)\tilde{s}}{n^-} \log \left(\frac{n^-}{(1-s)\tilde{s}} \right) \leq \frac{\log n^-}{n^-}$$

and

$$\frac{(1-s)(1-\tilde{s})}{n'} \log \left(\frac{n'}{(1-s)(1-\tilde{s})} \right) \leq \frac{\log n'}{n'}.$$

Then, we have

$$\begin{aligned} & \max \left\{ \frac{s \log \left(\frac{n}{s} \right)}{n}, \frac{(1-s)\tilde{s}}{n^-} \log \left(\frac{n^-}{(1-s)\tilde{s}} \right), \frac{(1-s)(1-\tilde{s})}{n'} \log \left(\frac{n'}{(1-s)(1-\tilde{s})} \right) \right\} \\ & \leq \max \left\{ \frac{\log n}{n}, \frac{\log n^-}{n^-}, \frac{\log n'}{n'} \right\} \\ & \leq \frac{\log(\min\{n, n^-, n'\})}{\min\{n, n^-, n'\}}. \end{aligned}$$

For brevity, we let $n_{\min} := \min\{n, n^-, n'\}$. It follows that with probability at least $1 - \delta$,

$$\begin{aligned} & \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \\ & \leq 2C_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + 3\ell^2 N \log(\max\{n, n^-, n'\} \ell N) + 6C_q \log \left(\frac{6}{\delta} \right) \right)^{\frac{q+1}{q+2}} \left(\frac{\log(n_{\min})}{n_{\min}} \right)^{\frac{q+1}{q+2}} \\ & \quad + 3C_{r,\alpha,d} 2^{q+1} N^{-\frac{\alpha(q+1)}{d}}. \end{aligned}$$

We choose $N \in \mathbb{N}$ to be the smallest integer satisfying

$$\left(N \frac{(\log(n_{\min}))^4}{n_{\min}} \right)^{\frac{1}{q+2}} \geq N^{-\frac{\alpha}{d}}, \quad (\text{F.22})$$

which is the balance of the two terms above: raising both sides to the power $q+1$ equates $(N(\log n_{\min})^4/n_{\min})^{\frac{q+1}{q+2}}$ with $N^{-\frac{\alpha(q+1)}{d}}$. That is,

$$N = \left\lceil \left(\frac{n_{\min}}{(\log(n_{\min}))^4} \right)^{\frac{d}{d+\alpha(q+2)}} \right\rceil. \quad (\text{F.23})$$

Then, for some $C'_{q,\alpha,d} > 0$, when $n_{\min} \geq C'_{q,\alpha,d}$, we have $\ell = (1 + \frac{\alpha}{d}) \frac{\log N}{\log 2} \leq N$ and thereby

$$\begin{aligned} \ell^2 N \log(\ell N) & \leq \left(2 \left(1 + \frac{\alpha}{d} \right) \frac{\log N}{\log 2} \right)^2 N \log N^2 \\ & \leq \frac{8}{(\log 2)^2} \left(1 + \frac{\alpha}{d} \right)^2 N (\log N)^3 \\ & \leq \frac{8}{(\log 2)^2} \left(1 + \frac{\alpha}{d} \right)^2 N (\log n_{\min})^3. \end{aligned}$$

Thus, with probability at least $1 - \delta$,

$$\begin{aligned} & \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \\ & \leq 2C''_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + (\log(\max\{n, n^-, n'\}))^{\frac{q+1}{q+2}} \right) \left(N \frac{(\log n_{\min})^3}{n_{\min}} \right)^{\frac{q+1}{q+2}} \\ & \quad + 3C_{r,\alpha,d} 2^{q+1} N^{-\frac{\alpha(q+1)}{d}}, \end{aligned}$$

where $C''_{q,\alpha,d}$ is a constant depending only on q, α, d .

By the choice of N at (F.23), we know

$$\left((N-1) \frac{(\log n_{\min})^4}{n_{\min}} \right)^{\frac{1}{q+2}} \leq (N-1)^{-\frac{\alpha}{d}}.$$

So, from $N \leq 2(N-1)$, the above estimate yields

$$\begin{aligned} & \varepsilon(f_{\text{ERM}}) - \varepsilon(f_c) \\ & \leq C''_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + (\log(\max\{n, n^-, n'\}))^{\frac{q+1}{q+2}} \right) 2^{\frac{q+1}{q+2}} (N-1)^{-\frac{\alpha(q+1)}{d}} + 3c_{r,\alpha,d} 2^{q+1} N^{-\frac{\alpha(q+1)}{d}} \\ & \leq \left(C''_{q,\alpha,d} \left(\log \left(\frac{6}{\delta} \right) + (\log(\max\{n, n^-, n'\}))^{\frac{q+1}{q+2}} \right) 2^{\frac{q+1}{q+2}} \cdot 2^{\frac{\alpha(q+1)}{d}} + 3c_{r,\alpha,d} 2^{q+1} \right) N^{-\frac{\alpha(q+1)}{d}} \\ & \leq \hat{C}_{q,\alpha,d,r} \left(\log \left(\frac{6}{\delta} \right) + \log(\max\{n, n^-, n'\}) \right) \left(\frac{n_{\min}}{(\log n_{\min})^4} \right)^{-\frac{\alpha(q+1)}{d+\alpha(q+2)}}, \end{aligned}$$

where $\hat{C}_{q,\alpha,d,r}$ is a constant depending on q, α, d, r .

Finally, by taking $\tau = N^{-\frac{\alpha}{d}}$, the restrictions $\tau \geq \max\{\frac{1}{n}, \frac{1}{n^-}, \frac{1}{n'}\}$ and $\tau \leq \min\{N2^{-\ell} + N^{-\frac{\alpha}{d}}, 1\}$ are both satisfied. This is because $\max\{\frac{1}{n}, \frac{1}{n^-}, \frac{1}{n'}\} = \frac{1}{n_{\min}}$ and

$$\tau = N^{-\frac{\alpha}{d}} \geq 2^{-\frac{\alpha}{d}} \left(\frac{n_{\min}}{(\log n_{\min})^4} \right)^{-\frac{\alpha}{d+\alpha(q+2)}} \geq 2^{-\frac{\alpha}{d}} (n_{\min})^{-\frac{\alpha}{d+\alpha(q+2)}} \geq 2^{-\frac{\alpha}{d}} \frac{1}{\sqrt{n_{\min}}}.$$

This concludes the proof.

G Experimental Details

We run experiments thrice and report the mean and standard deviation in Tables 3a, 3b, 4 and 5. The sweep tables of Section ?? are single runs and carry no dispersion, as that subsection states. Randomness is based on model initialization and data draws (for train/validation splits and synthetic anomalies).

We proceed to provide details on other experimental results for reproducibility and further discussion.

G.1 Per-Dataset Discussion of the AUROC Results

The AUROC results themselves are in the main text, in Table 4 alongside AUPR; this subsection gives the per-dataset reading of them.

On the NSL-KDD dataset, for known anomalies, VC-SA performs slightly worse than VC, while for unknown anomalies, VC-SA shows substantial improvements, especially as VC falls below random performance on DoS and probe attacks. On the Thyroid dataset, VC-SA outperforms VC for both known and unknown anomalies, whereas on the Arrhythmia dataset, VC-SA achieves roughly the same performance as VC for unknown anomalies.

For the MVTec dataset, VC-SA outperforms VC on 11 of the 14 categories in AUPR and 12 of 14 in AUROC, tying on Leather and losing only on Hazelnut and Metal Nut. The AUPR values for VC-SA are extremely close to 1.0 in nearly every category (e.g., Bottle: 0.997 vs. 0.996, Grid: 1.000 vs. 0.999), and AUROC shows a similar trend, with VC-SA achieving near-perfect accuracy

(e.g., Bottle: 0.994 vs. 0.991, Zipper: 0.993 vs. 0.985). This demonstrates that adding synthetic anomalies (SA) substantially boosts the model’s ability to detect true anomalies accurately.

For the AdvBench dataset, the AUPR values show a varied performance between VC and VC-SA across anomaly types, but the trend in AUROC mostly mirrors the AUPR results. For example, in the case of satnews, both AUPR (0.232 vs. 0.798) and AUROC (0.793 vs. 0.977) follow the same pattern where VC outperforms VC-SA. Similarly, for LUN, the VC model has higher AUPR (0.762 vs. 0.532) and AUROC (0.972 vs. 0.918). Overall, while performance differences exist, AUROC trends generally align with AUPR outcomes in AdvBench.

G.2 Datasets

Here, we present more details on the datasets we evaluate on. All numerical variables are normalized to be within 0 and 1 during training. For categorical variables, we one-hot encode them and sample from this discrete space (rather than in continuous space). During testing, if a datum contains a variable that is out-of-domain (i.e., not within 0 and 1 after normalization for numerical variables, or an unseen category for categorical variables), we can automatically assign the datum as anomalous. We do not encounter this trivial case during our experiments, so do not discuss this further. Details of each dataset are summarized in Table 6.

Table 6: Dataset details. Dimension is after one-hot encoding or feature extraction. Number of training data is rounded to one or two significant figures. The anomaly ratio represents the proportion of anomalies relative to the total number of training samples. We evaluated a range of anomaly ratios across datasets, including approximately 0.0008 (KDD), 0.05 (Thyroid), 0.23 (Arrhythmia), 0.09 (MVTec), and 0.59 (AdvBench).

Dataset	Data Type	Domain	Dimension	Num. Training Data	Anomaly Ratio
NSL-KDD	Tabular	Cybersecurity	119	70,000	0.0008
Thyroid	Tabular	Medical	21	4,000	0.05
Arrhythmia	Tabular	Medical	279	300	0.23
MVTec	Image	Manufacturing	1024	200-400	0.09
AdvBench	Language	Harmful Text	384	100,000	0.59

G.2.1 Tabular Data

NSL-KDD³, our cybersecurity benchmark dataset, has benign (normal) network traffic and 4 types of attacks (anomalies) during training and testing: Denial of Service (DoS), probe, remote access (RA), and privilege escalation (PE). To simulate semi-supervised AD, we use PE as known anomalies and the other 3 as unknown anomalies. We choose PE because it has the fewest training anomalies, allowing us to stress test our method.

Thyroid⁴ is a medical dataset with patient vitals. Besides normal data, there are also hyperfunction and subnormal anomalies. We select subnormal thyroid patient vitals to be known anomalies, while hyperfunction thyroid patient vitals are left as the unknown anomalies.

Arrhythmia⁵ is a medical dataset with patient vitals. There are patients with normal and various kinds of anomalous heartbeats. We chose left and right BBB as our known anomalies, and the rest

³Licensed to “redistribute, republish, and mirror” with reference to Tavallae et al. [49].

⁴CC-BY 4.0 license.

⁵CC-BY 4.0 license.

as unknown anomalies. We split the data into train and test datasets with a 0.2 test split. This dataset is small (see Table 6), so we make a few workarounds. First, for missing data, instead of dropping data, we use mean imputation, a common imputation method (see Xiao and Fan [53]). Second, comparing unknown anomaly categories with few anomalies can have much noise and may not be meaningful, so we group all unknown anomaly categories into 1 “unknown anomaly” category during evaluation. Third, we use all known anomalies during training, so we do not evaluate on known anomalies for testing.

G.2.2 Image Data

MVTec⁶ is an industrial manufacturing image dataset. It contains images of manufacturing items. These images include mostly properly manufactured products (i.e., normal images) and products with defects (i.e., anomalous images) that are grouped by the type of defect. To simulate semi-supervised AD, we include a type of defect along with the original working product for 14 different items. For standardization, we pick the first defect type according to alphabetical order. Due to the small dataset size (see Table 6), once again, we use all known anomalies during training and exclude known anomaly evaluation during testing. To convert image data to tabular form, we use 1024-dimensional DINOv2⁷ [34] (frozen) embeddings, like in Lau et al. [27]. We use these frozen embeddings rather than fine-tuning the feature extractor to avoid overfitting.

G.2.3 Language Data

AdvBench⁸ is a union of 10 smaller text datasets. Each data sample in the dataset corresponds to a sentence, and with it, a corresponding binary label of harmless (i.e., “normal”) or harmful (i.e., “anomalous”). There are 5 types of harms (anomalies): misinformation, disinformation, toxic, spam and sensitive, each of which describes 2 of the datasets (for a total of 10 datasets). To convert text into tabular form, we use 384-dimensional BERT⁹ [38] (frozen) sentence embeddings. Comments on using language models zero-shot for AD can be found in Yang et al. [55]. As a remark, this language dataset is by far the most different from the original tabular setting we described. Language comprises discrete tokens that are strung together sequentially, so representing it in tabular form is not a trivial task (let alone, a representation that is compact). Furthermore, defining what is harmful (i.e., anomalous) is highly contextual. For instance, a piece of misinformation may not be toxic, and a toxic text may not be misinformation. Due to this under-constrained nature of detecting harm, we use normal data from all datasets and harmful texts from 1 dataset from each category during training.

G.3 Compute Resources

To enhance reproducibility, we run experiments that a standard consumer-grade workstation can run. We use 128 GB of memory. We run neural networks on a single NVIDIA GeForce RTX 4090 GPU. Each experimental run takes at most one day to run, but usually within one hour. To run all experiments (including those without reported results), fewer than 10 days of active compute is estimated to be needed.

⁶CC BY-NC-SA 4.0 license.

⁷Apache License.

⁸Dataset has no license, provided by a Google drive link on <https://github.com/thunlp/Advbench>. Paper is published in EMNLP, which has a CC-BY 4.0 license.

⁹Apache License.

G.4 Composite Methods

Composite methods first do unsupervised AD to identify data that belong to the training classes, and then binary classifiers differentiate normal from known anomalous data given that the data are known. Concretely, we obtain a score from unsupervised AD on $P(\mathbf{x} \text{ belongs to training class})$ and a score from binary classifiers on $P(\mathbf{x} \text{ normal} | \mathbf{x} \text{ belongs to training class})$. Then, we can obtain a score of $P(\mathbf{x} \text{ normal}) = \mathbb{P}(\mathbf{x} \text{ from training class})P(\mathbf{x} \text{ normal} | \mathbf{x} \text{ from training class})$. We test kernel-based, tree-based and neural network methods: for unsupervised AD, we use OCSVM [43], Isolation Forest [31] and DeepSVDD [39]; for binary classifiers, we use SVM [10], Random Forests¹⁰ [22, 5] and neural networks. We multiply the score (shifted to be positive) from each unsupervised AD method with each binary classifier for a composite model (e.g., OCSVM with Random Forest), which produces $3 \times 3 = 9$ types of composite models. One limitation is that some models (e.g., SVM models, Isolation Forest) predict scores which are not probabilities. Nevertheless, we can still use their score to preserve the ranking of anomaly scores. The area under the precision-recall curve (AUPR) metric we use is a threshold-agnostic metric that allows us to evaluate the separability of anomalies from normal data in the output space. Although this two-stage approach is logical, it leads to an undesirable model: normal and known anomalies are modeled as similar, while known and unknown anomalies are modeled as different by unsupervised AD. Ideally, we would like to model the problem as how the problem arises: the normal class is different from all the anomalies [40], while anomalies (known or unknown) may or may not be similar [41].

Composite Models Are as Poor as Random For brevity, we omit the results of composite models because they all have random performance for both known and unknown anomalies in both NSL-KDD and AdvBench datasets, which we used for preliminary experiments. Although they were designed to inherit the benefits of both unsupervised AD and binary classifiers, they instead inherited errors from both.

In addition, to evaluate the reasonability of our method of composing scores, we ablated against different ways of composing scores: (1) normalizing scores to within 0 and 1 (which produces a score that resembles a probability score) and (2) normalizing scores by their sample variance. However, these other approaches to compose the output of models still produce random results. Hence, the straightforward approach of composing scores from unsupervised AD and binary classifiers does not seem like the right step forward. It is not unreasonable to believe that the composite models perform badly because the anomaly scores of different models are fundamentally different. The difference could lead to an inability to combine the scores, or that the scores are not cohesive due to a lack of integrated optimization.

G.5 Our Method

The full details can be found in our code implementation, which uses PyTorch for neural network implementation. Section 4 outlines our training process, with a visualization on the right of Figure 2. To provide greater clarity, we proceed to be more explicit in how we implement our method.

We set up a binary classification task between the normal and anomaly class. For training, the normal class has normal data, while the anomaly class has known and synthetic anomalies. Before training starts, synthetic anomalies are sampled uniformly from the data support $\mathcal{X}$. Specifically, we normalize all features to be between 0 and 1 and sample synthetic anomalies from $[0, 1]^d$ (note that for one-hot encoded variables, we sample from the discrete support instead). The base model is a

¹⁰We train SVM and Random Forest with balanced class weights.

ReLU network that predicts a scalar from 0 to 1¹¹ that describes how anomalous each datum is. Model weights are optimized via gradient descent on the loss.

During testing, the trained network predicts a scalar from 0 to 1 for normal data, known and unknown anomalies.

In the following two subsections, we outline two kinds of hyperparameter choices: how we represented some theoretical details that are impractical, and other key hyperparameters which may have differed from the theoretical description.

G.5.1 Implementing Theory

Our theoretical model in Definition 4.9 has sparsity constraints and requires all absolute values to be bounded by 1. These constraints are difficult to model during optimization. Hence, we use weight decay during gradient descent to model the preference toward sparser and smaller solutions.

Without knowledge of parameters like α , we do not have knowledge on what to set depth L^* and width w^* to. In particular, for width, our theoretical model seeks a width of $6(d + \lceil \alpha \rceil)N$. However, N can be $\mathcal{O}(e^d)$, which makes implementation computationally infeasible (e.g., $d = 119$ for NSL-KDD, so $N > 10^{51}$ and the width will be too large for training). Instead, in our experiments, we used the dimension of the input d and the number of data samples M to guide our choice of the width — higher dimension and more data should have wider networks for expressivity. We chose widths of 678 for NSL-KDD ($d = 119$, $M \approx 70,000$), 200 for Thyroid ($d = 21$, $M \approx 4,000$) and 500 for Arrhythmia ($d = 279$, $M \approx 300$), 6000 for MVTec ($d = 1024$, $M \approx 300$) and 6000 for AdvBench ($d = 384$, $M \approx 100,000$). We comment on depth in the next subsection.

G.5.2 Other Hyperparameters

In our initial experiments, we observe that the network does not converge during training and remains at high loss. We identify this as a symptom of vanishing (or zero) gradient. To alleviate this symptom, we make the following 3 changes. First, we make our network shallower, similar to the idea of structural risk minimization; our theoretical model seeks $L^* = 8 + (\ell + 5)(1 + \lceil \log_2(\max(d, \alpha)) \rceil)$ hidden layers, which is at least $8 + 6 \cdot 1 = 14$ for any d and equals 56 at $d = 119$, but we instead use *two* hidden layers. Depth is counted in hidden layers throughout, as in Definition 3.2, so two hidden layers means $L = 2$ in Table 5 and three weight matrices in code ($d \rightarrow w \rightarrow w \rightarrow 1$). The neural network depth is the main hyperparameter we tune. Second, we change out ReLU activations with leaky ReLU activations to avoid dead neuron problems. Third, instead of hinge loss, we use logistic loss to avoid zero loss on some samples. Models are then trained with early stopping on the validation loss.

These changes help with learning. We believe that the first change is the most significant, but keep the second and third change as well. This flexibility of switching out hyperparameters shows that our theory is not bound by specific hyperparameter choices and can be applied in general to standard binary classifiers.

For most datasets, the number of synthetic anomalies we use is $n' = n + n^-$, which equals the number of real training data points (i.e., normal data and known anomalies).¹² Adding more synthetic anomalies is still in line with Theorem 4.11, but we would like to avoid adding too many to avoid diluting the known anomaly supervision signal and contaminating the normal data during training. For AdvBench, we notice that validation performance is not perfect, denoting that some

¹¹In our theoretical model, the output is from -1 to 1, but the output can always be scaled according to match user preferences.

¹² n^- is usually small so $r \approx n$.

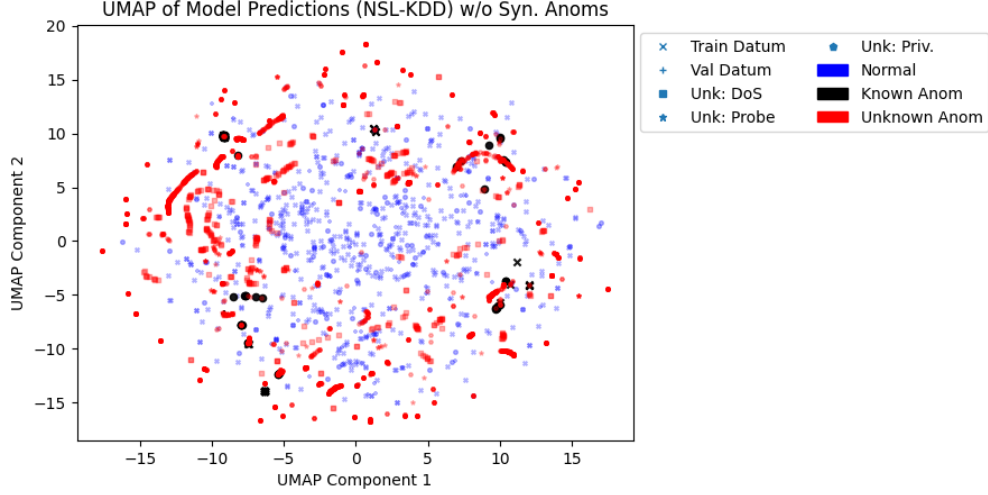


Figure 8: **Data visualization** of the model trained without synthetic anomalies. Unknown anomalies are more scattered than the model trained with synthetic anomalies.

anomalies are probably difficult. To avoid overfitting to known anomalies, we further increase the number of synthetic anomalies to $n' = 3 \cdot (n + n^-)$.

G.6 Are Synthetic Anomalies Representative of Unknown Anomalies?

Distributionally, the answer is no — our synthetic anomalies are drawn from a uniform distribution, while unknown anomalies can be drawn from any arbitrary distribution (which is likely more concentrated than the uniform distribution). In other words, good performance on synthetic anomalies does not necessitate good performance on unknown anomalies. The extreme case is when unknown anomalies are adversarial samples deliberately crafted to evade detection. In general, though, we observe that the model does not perform as well on unknown anomalies as it does on synthetic anomalies, even if these unknown anomalies are not deliberately crafted to be adversarial. Our validation AUPR with normal data, known and synthetic anomalies is always high (usually perfect) but AUPR on unknown anomalies can be low (e.g., models generally struggle on AdvBench dataset).

Nevertheless, our goal is not to create synthetic anomalies that resemble unknown anomalies — if unknown anomalies were known, we could directly do binary classification. Rather, we build on density level set estimation to detect unknown anomalies (Section 3). This is why we use validation loss and not validation performance for early stopping. Geometrically, we hope that synthetic anomalies can label the unknown regions / “open space” (i.e., regions without data) as anomalous, so models will be trained to classify those regions as anomalous. Then, when unknown anomalies appear in these regions, the model will be trained to identify these unknown anomalies as anomalous. The trade-off is that, a priori, we are unaware of which regions these unknown anomalies will appear in, so we will need to sample many synthetic anomalies to fill up more of these unknown regions (especially in higher dimensions), and hence will also require more model expressivity. To illustrate, in our data visualization in Figure 3, synthetic anomalies are scattered everywhere, some of them in regions (left of image) where real data are absent. Only a portion of synthetic anomalies are around the region of unknown anomalies (right of image).

Nonetheless, we observe that unknown anomalies cluster (e.g., bottom left, top of Figure 3), some quite closely to known anomalies. This clustering is not found in the model trained without

synthetic anomalies (Figure 8). Here, our model has learned discriminative features that are similar across real (known and unknown) anomalies.